

Fourfold nonvanishing by minimal metrics and moving jets

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Abstract

We prove canonical nonvanishing for smooth connected projective complex fourfolds: if K_X is pseudo-effective, then $H^0(X, mK_X) \neq 0$ for some positive integer m .

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1 Introduction

For a smooth projective variety X , canonical nonvanishing asks whether pseudo-effectivity of K_X guarantees a pluricanonical form: a nonzero section of mK_X for some integer $m > 0$. Pseudo-effectivity says that the numerical class is a limit of effective divisor classes. The question concerns the canonical line bundle itself, rather than an approximation to its numerical class. Over $\mathbb{C}$, pseudo-effectivity of K_X is equivalent to X not being covered by rational curves [5, Corollary 0.3]. Nonvanishing thus asks for a pluricanonical form on every smooth projective variety outside the uniruled class.

Nonvanishing is weaker than abundance. For a nef canonical divisor, abundance asks for *semiample*ness: some positive multiple is generated by its global sections and therefore defines a morphism. Obtaining the first section is a separate difficulty. We prove the following nonvanishing theorem in dimension four.

Theorem 1.1. *Let X be a smooth connected projective complex fourfold. If K_X is pseudo-effective, then $H^0(X, mK_X) \neq 0$ for some positive integer m .*

The theorem imposes no condition on numerical dimension, irregularity, or holomorphic Euler characteristic. Its conclusion is qualitative; it gives no uniform bound on the integer m .

The threefold results explain both the history of the problem and the lower-dimensional inputs used here. Miyaoka proved nonvanishing for minimal threefolds [34] and then abundance in numerical dimension one [35, pp. 203–204]. Kawamata’s work on pluricanonical systems and his abundance theorem for minimal threefolds established the broader canonical case [22, 23]. Keel–Matsuki–McKernan proved log abundance for threefolds, with the correction included here [25, 26]. Boundary arguments also require sections to agree where components meet: Fujino’s semi-log-canonical threefold abundance theorem supplies generation on the whole reduced floor, including its conductor identifications [10, Corollary 4.10]. These are established lower-dimensional theorems used before the new fourfold argument.

Several important fourfold cases were already known. Fujino proved semiampleness for canonical fourfolds with nef canonical divisor and positive irregularity. He distinguished nonvanishing for non-uniruled fourfolds of irregularity zero from the remaining abundance question in Kodaira dimension zero [11, Corollary 4.7 and Problems 4.10–4.11]. Lazić–Petrernell proved nonvanishing for $\mathbb{Q}$ -factorial terminal projective minimal varieties with canonical numerical dimension one and nonzero holomorphic Euler characteristic [30, Theorem 6.7]. Ambro’s canonical bundle formula gives a complementary reduction: the lower-dimensional minimal model program and abundance settle a nef klt adjoint whose nef reduction has lower-dimensional image [1, Theorem 4.3]. The nef reduction contracts the curves of degree zero through very general points; the dimension of its image is the *nef dimension*. This curve-theoretic dimension is distinct from numerical dimension, which is defined by nonzero intersection powers of the nef divisor.

The smooth theorem also has a log canonical consequence. Hashizume proved that smooth canonical nonvanishing in dimension n implies nonvanishing for log canonical pairs with real boundaries, as well as log minimal models, in dimensions at most n [20, Theorem 1.4]. For rational data we replace the resulting effective real representative by a rational one, preserving its support and working on the original variety.

Corollary 1.2. *Let (X, Δ) be a connected normal projective log canonical complex pair of dimension at most four. Suppose that $\Delta \geq 0$ is rational and $D = K_X + \Delta$ is $\mathbb{Q}$ -Cartier and nef. For every positive integer r such that rD is Cartier, there is a positive integer m with $H^0(X, \mathcal{O}_X(mrD)) \neq 0$.*

The corollary does not require X to be $\mathbb{Q}$ -factorial, and m may depend on the pair and the prescribed index. Log abundance asks for semiample of the nef adjoint $K_X + \Delta$. The passage from the first section to that conclusion uses the supported-boundary lifting and abundance-after-nonvanishing theorems in the separate companion *Lifting sections from the reduced support of an adjoint* [39, Theorems 1.1–1.2]. Together with the corollary, they give fourfold log abundance; that consequence does not enter the proof below.

A recent comparison is Liu–Xu’s preprint on good minimal models for log canonical pairs of dimension at most five with nonnegative invariant Iitaka dimension and numerical dimension at most one [31, Theorem 5.1]. Its hypotheses already include nonvanishing and restrict numerical dimension. The present smooth theorem obtains the first section without either restriction.

Geometry of a possible counterexample

Suppose nonvanishing fails. Existence of minimal models in dimension four allows us to fix a terminal minimal model Y , retaining the pluricanonical section spaces. The termination input used here is the ordinary-pair case of Chen–Tsakanikas [7, Theorem 1.1]; it supplies a minimal model, without asserting semiample. Write $K = K_Y$. The divisor K is nef and not big. The preceding reductions give $q(Y) = h^1(Y, \mathcal{O}_Y) = 0$ and full nef dimension. On a fixed very general locus, every curve has positive K -degree and every proper positive-dimensional subvariety is of general type. A fixed Cartier index makes the positive curve degrees uniformly bounded away from zero.

Choose ample polarizations approaching the nef ray of K , scaled so that their volumes remain small while their degrees on curves through that locus tend to infinity. A section with unusually high vanishing at a moving point would force a persistent component in the base locus of the corresponding jet systems. Differentiation with respect to the moving point controls the multiplicity along that component. This uses the parameter-differentiation method of Ein–Küchle–Lazarsfeld [9, Proposition 2.3]. We apply it to the whole space of sections satisfying the jet conditions, so the estimate controls ordinary powers of the ideal of the component. Intersection theory bounds its degree and canonical intersection. General type then produces a curve of bounded degree, contradicting the chosen scaling. The resulting upper bound applies to every section in every allowed Cartier degree. The independent moving-center estimates and the finite-cover lemma used in this argument and its two-factor extension are [38, Lemmas 7.1–7.5]. We apply them to the fourfold geometry and prove the required curve and correspondence obstructions here.

A second use of this argument concerns sections over two copies of Y . Fix an integer $\iota > 0$ making $S_0 = \iota K$ Cartier and consider

$$Z = \mathbb{P}(\mathcal{O}_{Y \times Y}(S_{0,1}) \oplus \mathcal{O}_{Y \times Y}(S_{0,2})) \longrightarrow Y \times Y.$$

A subscript indicates pullback from the corresponding factor. The two summands give two distinguished sections of this projective bundle. The symmetric-power formula records the distributions of a fixed tensor exponent between the two factors: its summands are $\mathcal{O}(aS_{0,1} + bS_{0,2})$, for nonnegative integers a, b with $a + b$ fixed. A polarization on Z combines the two base polarizations and a multiple of its tautological bundle. We show that it generates many jets at a very general point away from the two distinguished sections. Its Seshadri constant, the infimum of degree divided by multiplicity over curves through the point, measures this local positivity.

Moving base components in Z have several possible images and fibers. One case requires an additional argument. Restrict to a slice where one base coordinate is fixed. A multisection different from the two distinguished sections meets them in divisors whose

difference represents a multiple of K . This is a *signed representative*: its coefficients may be positive or negative. We must prove that the logarithmic adjoint on a resolution of its full support is big. This is where minimal metrics and ordinary cohomological injectivity enter the geometry.

The companion *Minimal metrics and interior injectivity for nef adjoints* [37] supplies two analytic inputs. Every minimal semipositive metric on the pullback of a nef projective klt adjoint has zero Lelong numbers. In addition, an injectivity theorem on ordinary H^1 compares an interior rational boundary with an endpoint boundary when the two endpoint bundles carry such metrics. The exact hypotheses are recorded in Section 4. These statements concern actual rational line bundles; numerical equivalence would not suffice.

The cohomological method draws on the harmonic-form approach of Enoki, as extended by Fujino to singular metrics, and on Matsumura’s treatment of metrics with transcendental singularities [15, Theorem 1.2, Lemmas 3.1–3.2, and Section 3, Claim 1] [32, Theorems 1.3 and 5.9, Sections 5.2–5.3]. Fujino’s complete metrics on an analytic complement and his comparison with coherent cohomology with multiplier ideals give a model for the setup. Matsumura supplies uniform estimates for primitives of exact forms.

Applied to two nearby nef klt adjoints, the cohomological injection makes restriction to a whole reduced non-klt boundary surjective. Lower-dimensional abundance and gluing across the conductor supply a section on that boundary. Once this section has been lifted, Gongyo–Matsumura’s fourfold criterion converts nonvanishing and the zero-Lelong metric into semiampleness [17, Corollary 5.3]. This order matters: the criterion is applied after the first section has been obtained. The signed-representative argument then excludes the offending multisection. The remaining analysis also excludes dominant finite correspondences, using a fixed branch complement and the positive canonical degree of curves.

From jets to a rank contradiction

We now have opposing estimates. Sections on one copy of Y can vanish only to a small order, while the projective bundle over $Y \times Y$ has enough jets to give a large evaluation rank. We fix the polarization, a smooth resolution and a flag of general hyperplane sections ending in a curve, together with one finite system of jets. The scalar estimate also supplies a fixed movable curve on the blowup of a point. Its intersection with effective divisors will bound orders of sections after reduction. We spread all these finite data and reduce modulo sufficiently large primes.

Ordinary jets produce an evaluation matrix whose generic rank is large. On the diagonal, the canonical Frobenius filtration expresses its possible values through truncated powers of the cotangent bundle. A single flag estimate bounds the sum of the dimensions of these spaces, forcing a much smaller rank there. A nonzero maximal determinant must therefore vanish to high order at a suitable diagonal point. Its two actual determinant line bundles, one from each factor, have classes controlled by the fixed movable curve. This bounds the order of every section of either factor on the reduction itself. The resulting small determinant order is incompatible with the order forced by the rank drop.

The filtration has a substantial history. For curves, the diagonal filtration appears in Raynaud’s work [40, Remarques 4.1.2(2)]; Joshi–Ramanan–Xia–Yu formulate it using the Cartier connection [21, Section 5.3]. Sun and Kitadai–Sumihiro give the higher-dimensional filtration and its local description by truncated monomials [41, Theorem 3.7] [27, Corollary 3.5]. Mustaŭ–Schwede’s comparison of ordinary and Frobenius powers provides the local jet conversion [36, proof of Proposition 2.12, Equation (2.8)], and Langer’s estimate controls the Frobenius instability error [28, Corollary 2.5]. The argument combines these tools with the two-factor geometry and the common flag estimate. Section 7 specifies the numerical

separation between the resulting ranks and orders and verifies the inputs of the common theorem.

The complete finite-data Frobenius comparison is proved independently in [38, Theorem 9.1]. It is also applicable to the all-dimensional abundance argument. Its hypotheses are the stated geometric data and numerical inequalities. No all-dimensional abundance conclusion or logarithmic Iitaka subadditivity is used in the fourfold preparation of these data.

Reading order. Section 2 fixes the terminal model and its curve-degree obstruction. Section 3 develops the moving-center tools and proves the scalar order bound. Section 4 states the analytic inputs, and Section 5 proves logarithmic general type for signed supports. Section 6 excludes the product’s moving centers and obtains the two-slot jets. Section 7 verifies the finite data for the common Frobenius theorem, completing smooth nonvanishing. Section 8 gives the original-variety and prescribed-index consequence.

Conventions. Varieties are over $\mathbb{C}$ except when the finite-data comparison is reduced to positive characteristic. A subvariety is of general type when a smooth projective resolution is of general type. Rational line bundles and their sections are interpreted after clearing specified denominators. An inequality in nef order means that its difference is nef. A very general locus is the complement of a countable union of proper closed subsets. The symbols r for a prescribed Cartier index, r_t for a scaling integer, $q(Y)$ for irregularity, and q for a projective-bundle weight have distinct uses.

2 A fixed minimal model of full nef dimension

Assume that smooth canonical nonvanishing fails in dimension four. We first choose a terminal minimal model on which every curve through a very general point has a fixed positive lower bound for its canonical degree. We then show that every proper positive-dimensional subvariety through such a point is of general type. These two properties will control the moving centers of the later jet argument. The model remains fixed while its ample perturbations vary.

Proposition 2.1. *If Theorem 1.1 fails, there is a projective $\mathbb{Q}$ -factorial terminal fourfold Y , with $K = K_Y$, such that*

- (i) K is nef, $\kappa(Y, K) = -\infty$, and $K^4 = 0$;
- (ii) $q(Y) = 0$ and the nef dimension of K is four;
- (iii) for some integer $\iota > 0$, ιK is Cartier and

$$K \cdot C \geq \frac{1}{\iota} \tag{2.1}$$

for every integral curve C through a point of a fixed very general locus in Y .

Proof. Start with a smooth counterexample H . Run the canonical minimal model program. Existence of the necessary flips follows from [3, Corollary 1.4.1]. Termination in the pseudo-effective fourfold case is [7, Theorem 1.1], applied with zero nef data to the ordinary pair with zero boundary. Divisorial contractions lower the Picard number, so together these statements give termination of the program. Pseudo-effectivity is preserved and excludes a Mori fiber space as its endpoint. We obtain a projective $\mathbb{Q}$ -factorial terminal minimal model Y .

Canonical section spaces in sufficiently divisible degrees are birationally invariant here. On a common resolution the differences from the respective pulled-back canonical divisors are effective exceptional divisors, and pushing their sheaves forward does not add sections. Thus $\kappa(Y, K) = -\infty$. In particular K is not big; as it is nef, this says $K^4 = 0$.

Terminal singularities are rational. Hence $q(Y)$ agrees with the irregularity of a resolution. If it were positive, [11, Corollary 4.7] would make K semiample, a contradiction. Likewise, if the nef dimension were at most three, [1, Theorem 4.3], with the established log minimal model program and log abundance in those dimensions, would make K semiample. This includes nef dimension zero. Thus the nef dimension is four.

The curve property of nef reduction [2, Theorem 2.1] says in this case that $K \cdot C > 0$ for every integral curve through a very general point. Choose ι with ι^*K Cartier. Its degree on an integral complete curve is an integer, so positivity gives Equation (2.1). This uses only the curve property of nef reduction, not an Iitaka fibration on Y . $\square$

Proposition 2.2. *For Y as in Proposition 2.1, every proper positive-dimensional subvariety through a point of a suitable fixed very general locus is of general type.*

Proof. We first work with one family of subvarieties, and at the end make the very general locus independent of the family. Hilbert schemes give countably many finite-type parameter spaces for integral subvarieties of Y . Stratification, resolution of the universal families, and generic smoothness give countably many smooth families

$$b : \mathcal{V} \longrightarrow B, \quad e : \mathcal{V} \longrightarrow Y,$$

whose fibers are smooth projective resolutions of their images and whose fiber maps are birational onto those images. To obtain this description, resolve the total universal family on an integral parameter stratum, shrink until its fibers are smooth and the fiber maps birational, and repeat on the complement by noetherian induction. Non-dominant evaluation families can be discarded by excluding their image closures in Y .

Consider a dominant family with fibers of dimension d , where $0 < d < 4$. We may slice the parameter space to dimension $4 - d$ so that e becomes generically finite and remains dominant. Indeed, the parameters incident to a general point of Y have a component of dimension $\dim B - (4 - d)$ on the locus where the fiber maps are generically embeddings. General ample cuts in a compactification of B meet this locus in finitely many points. We take the cuts very generally, so a very general point of the slice has the generic value of every plurigenus of the original family. There are only countably many such conditions.

On the resulting smooth total space, ramification and terminality give

$$K_{\mathcal{V}} \sim_{\mathbb{Q}} e^*K + E_{\mathcal{V}}, \quad E_{\mathcal{V}} \geq 0.$$

For completeness, the rational lift of e to a resolution of Y is defined at codimension-one points by properness. The usual ramification formula there, followed by the effective discrepancies of the terminal target, proves this divisor inequality. Its restriction to a very general fiber is

$$K_V \sim_{\mathbb{Q}} e_V^*K + E_V, \quad E_V \geq 0. \tag{2.2}$$

In particular K_V is pseudo-effective.

The dimension of V is at most three. The minimal model program and abundance in these dimensions give a good terminal minimal model $V_{\min}$. Suppose that V is not of general type. The semiample canonical divisor of $V_{\min}$ defines a morphism with connected positive-dimensional general fibers, on each of which a multiple of $K_{V_{\min}}$ is trivial. On a common resolution $\tilde{V}$ of V and $V_{\min}$, there is a complete curve through a very general

point with $K_{\tilde{V}}$ -degree zero. Here is the avoidance needed for this assertion. The singular locus of the terminal model and the centers of the exceptional divisors have codimension at least two. A general fiber either misses a given vertical center or meets a dominant center in codimension at least two. General sufficiently ample complete intersections in that fiber through a prescribed very general point give a curve avoiding all such centers; for a one-dimensional fiber the fiber itself does so. Its strict transform has canonical degree zero by the exceptional discrepancy formula.

Pulling back Equation (2.2) to $\tilde{V}$ and adding the effective discrepancies of $\tilde{V} \rightarrow V$ gives

$$K_{\tilde{V}} \sim_{\mathbb{Q}} \tilde{e}^* K + \tilde{E}, \quad \tilde{E} \geq 0.$$

Choose the preceding curve through a point outside $\text{Supp } \tilde{E}$ and outside the exceptional locus of the birational map onto the image subvariety. It is not contained in $\tilde{E}$, and its image in Y is a curve. Thus its pulled-back K -degree is at most zero. Dominance of the total evaluation lets us choose the point in the very general locus of Equation (2.1), which gives a contradiction.

We have proved that the very general fiber of the sliced family is of general type. By the choice of slice its plurigenera are the generic plurigenera of the original smooth family. Upper semicontinuity shows that all fibers of that family have at least these plurigenera, hence are of general type as well. Finally exclude the image closures of all non-dominant families. Countability of the parameter strata gives the asserted very general locus. $\square$

For the rest of the contradiction argument, fix Y , K , ι , and a very general locus on which both propositions apply. We next choose ample perturbations of K with bounded volume and growing degree on every curve through this locus. The two propositions will then turn a section of excessive vanishing order into a contradiction.

3 Moving centers and scalar vanishing

We work on the terminal fourfold Y from Section 2, write $K = K_Y$, and put $n = 4$. Write Y^{vg} for a fixed very general locus on which both Propositions 2.1 and 2.2 apply. Thus every curve through this locus has K -degree at least $1/\iota$, and every proper positive-dimensional subvariety through it is of general type. We will bound the order of every section of every Cartier multiple of a suitable ample rational divisor at a very general point. Fix a very ample Cartier divisor A on Y . For a fixed positive rational ϵ and positive rational t tending to zero, define

$$\mu_t = ((K + tA)^n)^{1/n}, \quad r_t \in \mathbb{Z}_{>0}, \quad r_t \mu_t \rightarrow \epsilon, \quad L = L_t = r_t(K + tA). \quad (3.1)$$

Such integers r_t exist, for example by rounding ϵ/μ_t . Since K is nef and not big, $\mu_t > 0$ tends to zero. Hence

$$r_t \rightarrow \infty, \quad L_t^n \rightarrow \epsilon^n, \quad L_t \cdot C \geq r_t/\iota \quad (3.2)$$

for every integral curve C through that very general locus. The integers r_t are unrelated to the original Cartier index r in Corollary 1.2.

A section with excessive vanishing will yield a curve through Y^{vg} whose L_t -degree is bounded independently of t , contradicting Equation (3.2). To construct that curve, we follow the base loci of the full jet kernels as their marked point moves. A component that persists at two successive jet levels has high multiplicity; this bounds both its degree and its canonical intersection. General type then produces a curve of bounded degree. We first give

the three moving-center estimates from [38, Lemmas 7.1–7.4], including the differentiation argument responsible for ordinary ideal powers. These inputs are independent of abundance and subadditivity. We then prove the curve obstruction and apply it to scalar vanishing.

3.1 Numerical subadjunction at a generic lc center

The numerical statement below only requires log canonicity near the generic point of the center. This is the form needed for boundaries obtained from a linear system.

The subadjunction method relates an lc center to the canonical bundle formula. Kawamata’s minimal-center theorem and the later unperturbed form of Fujino–Gongyo are important antecedents [24, 16]. The numerical form below permits a center that is only generically lc and need not be minimal. Its proof in the companion uses a dlt stratum, the generic rank-one condition, b-nef moduli, and finite-base ramification; it is not an application of those antecedents with their hypotheses omitted.

Lemma 3.1 (Numerical generic subadjunction). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety over $\mathbb{C}$, let $\Theta \geq 0$ be a rational divisor, and let $V \subset Z$ be an integral positive-dimensional subvariety. Suppose that (Z, Θ) is lc at the generic point of V and that a divisor of log discrepancy zero has center V . Let $f = \dim V$, let P be a nef rational Cartier divisor on Z , and let $\rho : V^* \rightarrow V$ be a smooth projective resolution, factoring through the normalization. Then*

$$K_{V^*} \cdot (\rho^* P)^{f-1} \leq (K_Z + \Theta)|_V \cdot (P|_V)^{f-1}. \quad (3.3)$$

The right side is the intersection of the restricted rational Cartier class with the cycle $[V]$. No $\mathbb{Q}$ -Gorenstein hypothesis on the normalization of V is required.

The proof is given in the independent moving-center part of [38, Lemma 7.1].

3.2 Degree and canonical bounds for a base component

The following companion lemma applies numerical subadjunction to a boundary built from an actual linear system. Its local multiplicity controls both the degree of the center and the coefficient of that boundary.

Lemma 3.2 (A base component of high multiplicity). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety of dimension d , let P be an ample rational Cartier divisor, and let $k > 0$ be an integer for which kP is Cartier. Let $0 \neq \mathcal{H} \subset H^0(Z, \mathcal{O}_Z(kP))$ be a linear subspace with proper base locus and base ideal $\mathfrak{b}$. Suppose that V is an integral base-locus component of dimension $f < d$, isolated at its generic point η_V , and that $\eta_V \in Z_{\text{sm}}$. Write $a = d - f$ and $\mathfrak{m} = \mathfrak{m}_{\mathcal{O}_{Z, \eta_V}}$. If*

$$\mathfrak{b}\mathcal{O}_{Z, \eta_V} \subset \mathfrak{m}^h \quad (h \in \mathbb{Z}_{>0}),$$

then

$$P^f \cdot V \leq (k/h)^a P^d. \quad (3.4)$$

If $f > 0$, there is a rational number c with $0 < c \leq ak/h$ such that, on a smooth resolution V^ ,*

$$K_{V^*} \cdot P^{f-1} \leq (K_Z + cP)|_V \cdot P^{f-1}. \quad (3.5)$$

The coefficient c may be chosen with an effective rational divisor $\Theta \sim_{\mathbb{Q}} cP$ such that (Z, Θ) is lc at the generic point of V and has an lc place with center V . Thus Lemma 3.1 can also be applied with a different nef testing class. The degree estimate includes $f = 0$; the canonical estimate is asserted only for positive-dimensional V .

The proof is given in the independent moving-center part of [38, Lemma 7.2].

3.3 Differentiating the full moving jet kernel

We use the parameter-differentiation method developed in [9, Proposition 2.3]. The form used here retains ordinary ideal powers and isolated base components. Fixing a high order at a varying point produces a family of linear subspaces of one fixed section space. A component common to two successive base loci acquires high multiplicity, because parameter derivatives of the higher kernel still belong to the lower kernel. We state both the parameter-family conclusion and the local slicing property that will be used later.

Lemma 3.3 (Moving multiplicity). *Let Z be an integral projective variety of dimension d , and let P be an ample rational Cartier divisor. Fix an integer k with kP Cartier and positive real numbers*

$$\tau_0 < \tau_1 < \cdots < \tau_d, \quad \tau_{i+1} - \tau_i = \delta > 0.$$

For a smooth point x , let $\mathcal{H}_i(x)$ be the entire subspace of sections of kP vanishing at x to order at least $\lceil k\tau_i \rceil$. Suppose that for very general x all these subspaces are nonzero and the base locus of $\mathcal{H}_0(x)$ has a positive-dimensional component through x . If $k\delta \geq 4$, then, after an algebraic parameter extension and restriction to nonempty opens, there are an irreducible parameter space T , a fixed adjacent pair of levels, and a family of integral subvarieties V_t common to the two base loci such that the incidence

$$\Gamma = \{(t, z) : z \in V_t\} \subset T \times Z$$

dominates Z . Each general V_t has dimension in $[1, d-1]$, is an isolated component of the higher base locus at its generic point, and its entire higher-series base ideal is contained there in

$$\mathcal{I}_{V_t}^h, \quad h = \lceil k\delta/2 \rceil, \quad k/h \leq 2/\delta. \quad (3.6)$$

These are ordinary local ideal powers in the smooth ambient open.

More precisely, the incidence base ideal lies in $\mathcal{I}_\Gamma^h$ on a dense open of Γ . If a further morphism $Z \rightarrow B$ is given, one may choose a general incidence point where the base support is only V_t and V_t is smooth over the smooth open of its image in B . For the fiber component F through that point, assume that F is proper in the integral ambient fiber component under consideration. The restricted series is then nonzero and has base ideal in the ordinary $\mathcal{I}_F^h$ and has F as an isolated base component there. This last assertion concerns the scheme fiber locally at that chosen point, and does not assert transversality of every fiber. Smoothness near the generic point of F , and the projectivity and klt hypotheses needed for Lemma 3.2 on the ambient fiber, must be checked in each application.

Proof. We recall the argument of [38, Lemmas 7.3–7.4] to explain why the estimate concerns the entire kernel and ordinary ideal powers. On the open where the jet maps have constant rank, their kernels are vector bundles with fibers $\mathcal{H}_i(x)$. Their base loci increase with i . Starting with a positive-dimensional component through the mark, follow a containing component at each level. The $d+1$ components all have dimensions in $[1, d-1]$, so two successive ones coincide. After a finite parameter extension and shrinking, their geometric generic components and inclusions spread to one family $\Gamma \subset T \times Z$. Its evaluation dominates Z , since each member contains its moving mark $x(t)$.

Take a local frame of the higher kernel and trivialize kP near z with a frame independent of t . In a fixed basis of the global section space, each frame section has the form

$$s(t, z) = \sum_{\alpha} c_{\alpha}(t) s_{\alpha}(z).$$

Differentiation in the parameter with z fixed therefore gives another global section of kP . A derivative of order r lowers the vanishing order at $x(t)$ by at most r . It belongs to the entire lower kernel whenever

$$r \leq \lceil k\tau_{i+1} \rceil - \lceil k\tau_i \rceil.$$

The right side is at least $k\delta - 1$, so every derivative of order less than $h = \lceil k\delta/2 \rceil$ belongs to that kernel and vanishes on Γ . This is where using the full kernels matters: an arbitrarily chosen subseries need not contain these derivatives.

To turn these vanishings into ordinary ideal membership, work at a general smooth incidence point. Since $\Gamma \rightarrow Z$ is submersive, parameter directions span its normal space in $T \times Z$. Choose parameter coordinates $t = (t', t'')$ so that the implicit function theorem writes Γ as $t' = F(t'', z)$. With t'' and z fixed, derivatives in t' are exactly derivatives in the normal coordinates $u = t' - F(t'', z)$. All normal Taylor coefficients of degree below h therefore vanish, proving $s \in \mathcal{I}_\Gamma^h$. Faithful flatness of analytic local rings over algebraic local rings gives the same membership algebraically. There are finitely many frame sections, so this containment holds for the entire incidence base ideal on a dense algebraic open.

Generic smoothness of $\Gamma \rightarrow T$ identifies its scheme fiber there with the reduced member V_t . Restriction gives $\mathcal{I}_\Gamma^h \mathcal{O}_{\{t\} \times Z} = \mathcal{I}_{V_t}^h$, and the frame specializes to a basis of the entire higher kernel. Isolation holds because V_t is a base-locus component. Finally remove all other base components and choose the incidence point where V_t is smooth over its actual image in B . Its scheme fiber is then reduced locally, so restriction to the ambient fiber sends $\mathcal{I}_{V_t}^h$ to $\mathcal{I}_F^h$ and the base support to F . If F is proper in the chosen integral ambient fiber component, that support is proper there; hence the restricted series is nonzero and has F as an isolated base component at its generic point. $\square$

For a fixed gap δ , combining Lemmas 3.2 and 3.3 gives degree and canonical-intersection bounds for a repeated moving component, with coefficients independent of the large divisible integer k . The next lemma converts such intersection bounds into a curve of bounded degree. Its general-type hypothesis supplies a birational linear system in a degree depending only on the dimension. The scalar application uses Proposition 2.2 and has no boundary. We include a reduced boundary in the statement because the two-slot argument will also need its logarithmic form.

3.4 A uniform obstruction from logarithmic general type

Lemma 3.4 (Bounded-degree curve obstruction). *Keep Y , L_t , and ι as in Equations (3.1) and (3.2). Fix $f \geq 1$ and positive constants C_0, C_1 . Suppose, for arbitrarily small t , that there are a smooth projective variety V_t^* of dimension f , a reduced simple normal crossings divisor G_t on it, a nef rational divisor P_t , and a morphism $e_t : V_t^* \rightarrow Y$ satisfying the following conditions:*

- (i) $K_{V_t^*} + G_t$ is big;
- (ii) $0 < P_t^f \leq C_0$ and $(K_{V_t^*} + G_t) \cdot P_t^{f-1} \leq C_1 P_t^f$;
- (iii) e_t is generically finite onto its positive-dimensional image, and that image meets the fixed very general locus of Y ;
- (iv) $P_t - e_t^* L_t$ is nef.

Then these conditions are impossible for all sufficiently small t . The case $G_t = 0$ includes ordinary general type.

Proof. Uniform effective birationality for projective lc pairs of dimension f with big adjoint and coefficients in the fixed DCC set $\{0, 1\}$ supplies an integer $b = b(f)$ such that $|b(K_{V_t^*} + G_t)|$ is birational [18, Theorem 4.0.1]. Resolve its base ideal and denote the free moving divisor by H_b . We keep the same notation for the pullbacks of P_t and L_t . The fixed divisor is effective, so

$$H_b \cdot P_t^{f-1} \leq b(K_{V_t^*} + G_t) \cdot P_t^{f-1} \leq bC_1 P_t^f.$$

Both H_b and P_t are nef and big; P_t is big because $P_t^f > 0$. The Khovanskii–Teissier log-concavity inequalities for their mixed intersections give, when $f > 1$,

$$P_t \cdot H_b^{f-1} \leq \frac{(P_t^{f-1} \cdot H_b)^{f-1}}{(P_t^f)^{f-2}} \leq (bC_1)^{f-1} P_t^f \leq (bC_1)^{f-1} C_0. \quad (3.7)$$

Indeed the successive ratios of the positive numbers $P_t^{f-i} H_b^i$ decrease, which gives the first inequality.

Choose a point where the birational morphism defined by H_b is an isomorphism onto an open of its image, where e_t is quasi-finite, and whose image lies in the very general locus of Y . These conditions are compatible. To spell out the last one, write the excluded subset of Y as a countable union of proper closed subsets. The image of e_t is not contained in any of them because it meets their complement. Their inverse images are therefore proper closed subsets of V_t^* . They can be avoided together with the finitely many dense-open conditions. For a sweeping family one first chooses a very general member not contained in any excluded subset, and then such a point on that member.

For $f > 1$, cut by $f - 1$ general members of $|H_b|$ passing through the chosen point. These give a proper effective curve cycle: on the birational image they are general hyperplanes through a point of the isomorphism open, and their pullbacks have no positive-dimensional base component. A component C through the selected point maps nonconstantly to Y , since e_t is quasi-finite there. Nefness and Equation (3.7) give

$$L_t \cdot e_t(C) \leq e_t^* L_t \cdot C \leq P_t \cdot C \leq P_t \cdot H_b^{f-1} \leq (bC_1)^{f-1} C_0.$$

In the first inequality the positive degree of the finite map of the normalization of C onto its image has merely been discarded. If $f = 1$, use V_t^* itself as C ; the same argument gives $L_t \cdot e_t(C) \leq P_t \cdot C \leq C_0$.

The image is an integral curve through the very general locus, so Equation (3.2) bounds its degree below by r_t/ι . This tends to infinity, whereas the upper bound depends only on f, C_0, C_1 . The contradiction proves the lemma. If several dimensions $1 \leq f \leq n$ occur, take the maximum of the finitely many resulting constants. $\square$

3.5 The scalar order bound

Define

$$P_s = 2r_t(K + 2tA), \quad \rho_t = (P_s^n)^{1/n}. \quad (3.8)$$

The inequalities

$$2L \leq P_s \leq 4L$$

are inequalities in nef order. After decreasing t , they imply

$$\epsilon \leq \rho_t \leq 8\epsilon. \quad (3.9)$$

Only the existence of a positive lower bound and the displayed upper bound will be used.

Proposition 3.5 (Scalar orders). *For all sufficiently small positive rational t , a very general point $x \in Y_{\text{sm}}$ has the following property. For every positive integer k for which kP_s is Cartier and every nonzero section $s \in H^0(Y, \mathcal{O}_Y(kP_s))$, one has*

$$\text{ord}_x(s) \leq 4k\rho_t \leq 32k\epsilon. \quad (3.10)$$

Proof. Fix t for the moment. For each Cartier multiple kP_s and each integer a , the condition

$$H^0(Y, \mathcal{O}_Y(kP_s) \otimes \mathcal{I}_x^a) \neq 0$$

is a rank condition on the jet evaluation map over Y_{sm} . We choose x outside all proper rank-jumping loci of these countably many maps. Thus if the order bound fails at such a point, the relevant high-order kernel is nonzero for general marks as well. Taking powers of the offending sections gives the same strict inequality in arbitrarily large divisible degrees. This permits all subsequent choices of a sufficiently large divisible k .

Choose $n + 1$ levels equally spaced from $2\rho_t$ to $3\rho_t$; their gap is

$$\delta_s = \rho_t/n.$$

If $\text{ord}_x(s) > 4k\rho_t$ in a sufficiently large degree, every one of these jet kernels is nonzero. The base locus of the lowest kernel cannot have x as an isolated component. Indeed its base ideal at x is contained in $\mathfrak{m}_x^{\lceil 2k\rho_t \rceil}$, so the zero-dimensional case of Lemma 3.2 would give

$$1 \leq \left(\frac{k}{\lceil 2k\rho_t \rceil} \right)^n P_s^n \leq 2^{-n},$$

a contradiction.

Apply Lemma 3.3. It produces a dominant family of proper positive-dimensional base components V , of some dimension $1 \leq f < n$, such that the entire higher jet kernel's generic base ideal along V is primary and contained in $\mathcal{I}_V^h$, where

$$h = \lceil k\delta_s/2 \rceil, \quad k/h \leq 2/\delta_s.$$

For a general member, Lemma 3.2 gives

$$P_s^f \cdot V \leq (2/\delta_s)^{n-f} P_s^n, \quad (3.11)$$

$$K_{V^*} P_s^{f-1} \leq \left(\frac{1}{2r_t} + \frac{2(n-f)}{\delta_s} \right) P_s^f \cdot V, \quad (3.12)$$

where V^* is a smooth projective resolution. The second inequality uses $K \leq P_s/(2r_t)$.

The constants on the right are bounded independently of small t , of k , and of the member V , by Equation (3.9). We choose a very general incidence point before choosing the member V . Its image belongs to Y^{vg} , so V^* is of general type by Proposition 2.2. Moreover P_s dominates L in nef order. The inclusion of V into Y , composed with its resolution, satisfies all the hypotheses of Lemma 3.4. Equations (3.11) to (3.12) would therefore give a curve through Y^{vg} of bounded L -degree, contradicting Equation (3.2) as t becomes small.

The bounds used in this contradiction do not depend on the initial degree in which excessive vanishing occurred. Consequently, after one restriction to sufficiently small t , all Cartier multiples satisfy Equation (3.10) at a very general point. $\square$

4 The analytic inputs and the lifting problem

The geometry will produce a divisor representing K_Y with coefficients of both signs. We shall prove that its full support is of logarithmic general type. The difficulty is to find the first section of an auxiliary nef adjoint before invoking a semiample criterion. The two analytic results below, proved in the complete companion [37], address exactly that difficulty. They concern actual rational line bundles.

A semipositive singular Hermitian metric has *minimal singularities* if its local weight is, up to a bounded additive error, no more singular than the weight of any other semipositive metric on the same bundle. For a rational line bundle this definition is made after taking one Cartier multiple and dividing the weights by that multiple.

Theorem 4.1 (Zero Lelong numbers for nef klt adjoints). *Let (H, Θ) be a normal connected projective complex klt pair, where $\Theta \geq 0$ is rational and $D_H = K_H + \Theta$ is nef and $\mathbb{Q}$ -Cartier. Let $\pi : V \rightarrow H$ be any projective log resolution. The rational line bundle $\pi^* D_H$ admits a semipositive metric with minimal singularities, and every such metric has zero Lelong number at every point of V . Its multiplier ideal is trivial for every positive exponent. Neither K_H nor Θ is required to be separately $\mathbb{Q}$ -Cartier.*

This is [37, Theorem 1.1]. The adjoint hypothesis is material: a general nef line bundle need not have this metric regularity [8, Example 1.7].

Theorem 4.2 (Ordinary interior injectivity). *Let V be a smooth projective complex variety and L an integral divisor. Let $0 \leq C_0 \leq C_2$ be effective rational divisors whose combined support has simple normal crossings. Suppose the actual rational line bundles $L - C_0$ and $L - C_2$ have semipositive singular Hermitian metrics with zero Lelong numbers at every point. For a rational number $0 < \lambda < 1$ put $C_1 = (1 - \lambda)C_0 + \lambda C_2$. The natural inclusion of sheaves induces an injection*

$$H^1(V, \mathcal{O}_V(K_V + L - \lfloor C_1 \rfloor)) \longrightarrow H^1(V, \mathcal{O}_V(K_V + L - \lfloor C_0 \rfloor)).$$

The complete proof, including its passage to ordinary coherent cohomology, is in [37, Theorem 1.2]. The analytic methods have antecedents in Fujino's singular-metric treatment of harmonic forms and Matsumura's uniform control of primitives [15, Theorem 1.2 and Lemmas 3.1–3.2] and [32, Theorems 1.3 and 5.9]. Those results are not being identified with the interior comparison.

Here is the lifting problem to which we will apply the theorem. For a nef rational adjoint N on a normal projective variety H and a reduced closed subscheme $S \subset H$, choose $m > 0$ with mN Cartier. The restriction sequence is

$$0 \longrightarrow \mathcal{O}_H(mN) \otimes \mathcal{I}_S \longrightarrow \mathcal{O}_H(mN) \longrightarrow \mathcal{O}_S(mN) \longrightarrow 0.$$

The obstruction to lifting a section on S is its image under the connecting homomorphism, whose image is the kernel of

$$H^1(H, \mathcal{O}_H(mN) \otimes \mathcal{I}_S) \longrightarrow H^1(H, \mathcal{O}_H(mN)).$$

Thus injectivity of this ordinary cohomology map suffices to lift every section on the whole scheme S . In the next section, S is the reduced non-klt floor of an auxiliary lc boundary. Two nearby nef klt adjoints supply the endpoint metrics, and multiplier-ideal local vanishing identifies the theorem's map with this inclusion. Threefold abundance supplies a section on the entire floor, with the conductor identifications already imposed.

5 Signed representatives and logarithmic general type

Fix the terminal canonical counterexample Y of Proposition 2.1, with the very general locus supplied by Proposition 2.2. The particular need for a signed divisor comes from the slice bundle $\mathbb{P}(\mathcal{O}_Y \oplus \mathcal{O}_Y(\iota K_Y))$. Its two summands determine disjoint sections. Intersecting an integral multisection of degree $e > 0$, distinct from those sections, with them and pushing to Y gives effective divisors D_0, D_∞ with $D_0 - D_\infty \sim e\iota K_Y$, as verified in Equation (6.25). Thus the representative of K_Y has signed coefficients, whereas the boundary is the reduced union of the two supports, including any components that cancel in their difference. We prove the logarithmic bigness needed to apply Lemma 3.4 to this boundary once its intersection degree has been bounded.

The proposition allows any reduced boundary containing the signed support. Its proof constructs an auxiliary nef adjoint with a nonempty reduced floor. The analytic results in Section 4 lift a section from that whole floor; only after this establishes nonnegative Kodaira dimension do we apply the required semiamplessness theorem.

Proposition 5.1 (Signed representative alternative). *Let Y be a projective $\mathbb{Q}$ -factorial terminal fourfold such that K_Y is nef and $\kappa(Y, K_Y) = -\infty$. Suppose that there is a very general locus of Y through whose points every proper positive-dimensional subvariety is of general type on resolution. Let J be a rational Weil divisor, with coefficients of either sign, satisfying $J \sim_{\mathbb{Q}} K_Y$. Let $p : W \rightarrow Y$ be a projective log resolution, and let D_W be a reduced simple normal crossings divisor containing the strict support of J and every p -exceptional divisor. Then $K_W + D_W$ is big.*

Proof. Put $P_W = K_W + D_W$. Terminality gives

$$K_W = p^* K_Y + E_W, \quad E_W \geq 0$$

with E_W exceptional. In particular, both K_W and P_W are pseudo-effective. We assume that P_W is not big and derive a contradiction. We first reduce its Kodaira dimension to at most zero, then construct a nef boundary with a nonempty reduced floor. We will lift a nonzero section from that entire floor before using any semiamplessness theorem that requires nonnegative Kodaira dimension.

Positive Kodaira dimension would already imply bigness

Suppose first that $\kappa(P_W) > 0$. Resolve the rational map defined by a moving subsystem of a sufficiently divisible multiple bP_W :

$$\pi : \widetilde{W} \longrightarrow W, \quad f : \widetilde{W} \longrightarrow U.$$

Here $\widetilde{W}$ is smooth, U is the projective image, and, for a very ample line bundle $\mathcal{O}_U(1)$,

$$b\pi^* P_W \sim f^* \mathcal{O}_U(1) + F, \quad F \geq 0. \quad (5.1)$$

If f is generically finite, then P_W is big, contrary to the assumption. Otherwise $0 < \dim U < 4$. A smooth fiber component through a very general point of $\widetilde{W}$ is birational to a proper positive-dimensional subvariety of Y meeting the very general locus. It is therefore of general type. We work over a smooth open of U where generic smoothness applies. On its smooth fibers the restriction of $K_{\widetilde{W}}$ is the canonical bundle of the fiber, up to the constant one-dimensional factor coming from the base.

Fix an ample Cartier divisor H_0 on $\widetilde{W}$. For some integer $j > 0$, the restriction of $jK_{\widetilde{W}} - H_0$ has a nonzero section on such a fiber component. Distinct components of a

smooth fiber are disjoint, so the section can be taken to be zero on the other components. Choose the fiber over a point where the fiber dimension of the space of sections is the generic one for every integer j ; this requires only countably many exclusions. Consequently

$$f_*\mathcal{O}_{\widetilde{W}}(jK_{\widetilde{W}} - H_0)$$

is generically nonzero for one j . After tensoring by a sufficiently large power $\mathcal{O}_U(c)$, this coherent sheaf has a nonzero global section. Thus

$$jK_{\widetilde{W}} + cf^*\mathcal{O}_U(1) - H_0 \sim E_0, \quad E_0 \geq 0.$$

The divisor on the left before subtracting H_0 is big. By Equation (5.1), adding an effective divisor gives bigness of $jK_{\widetilde{W}} + cb\pi^*P_W$. Birational pushforward gives bigness of $jK_W + cbP_W$. For completeness, the ample divisor H_0 has big pushforward: for a fixed ample divisor H_W on W and small positive rational η , $H_0 - \eta\pi^*H_W$ is ample, and pushing an effective rational representative proves this assertion. Finally

$$(j + cb)P_W = jK_W + cbP_W + jD_W$$

is big. This contradiction proves

$$\kappa(P_W) \leq 0. \tag{5.2}$$

A minimal model and a crepant klt perturbation

Run a log minimal model program for the rational dlt pair (W, D_W) . We use flip existence for klt pairs, the usual dlt contraction properties, and termination of flips for pseudo-effective lc fourfolds, applied with zero nef part [3, 7]. The dlt flips needed here are obtained by lowering the finitely many boundary coefficients slightly while retaining negativity of the given extremal ray. The resulting pair is klt. On the relative Picard-number-one contraction, the original and perturbed negative adjoints are relatively proportional with positive rational ratio; the contraction theorem descends a Cartier multiple of their numerically trivial difference. The klt flip therefore has the required positivity for the original adjoint as well. This supplies the flips used in this dlt program.

There are only finitely many divisorial contractions. Pseudo-effectivity persists under each pushforward and strict transform and excludes a negative fiber space, by intersection with covering curves in a general fiber. These pushforwards on numerical divisor classes are well-defined on the $\mathbb{Q}$ -factorial models: on a common resolution they are pullback followed by pushforward, and exceptional divisor classes give no ambiguity. We obtain a projective $\mathbb{Q}$ -factorial dlt minimal model (Z, D_Z) without extracting divisors. Write

$$M = K_Z + D_Z.$$

The divisor D_Z is reduced, M is nef, and the standard exceptional comparison on a common resolution preserves section spaces in sufficiently divisible degrees. Hence

$$\kappa(M) = \kappa(P_W) \leq 0.$$

Moreover K_Z is pseudo-effective, since K_W is pseudo-effective.

The signed representative supplies an additional linear relation. Indeed

$$P_W \sim_{\mathbb{Q}} p^*J + E_W + D_W,$$

and the divisor on the right is supported on D_W . Its strict pushforward to Z is a signed rational divisor G_Z supported on D_Z , with

$$M \sim_{\mathbb{Q}} G_Z. \quad (5.3)$$

We next make a small klt perturbation while keeping a fixed Cartier multiple of M . Choose an integer $\ell > 0$ for which ℓM is Cartier, and fix a rational number

$$0 < \delta < \min \left\{ \frac{1}{2}, \frac{1}{16\ell + 2} \right\}. \quad (5.4)$$

The klt adjoint

$$M - \delta D_Z = K_Z + (1 - \delta)D_Z$$

is pseudo-effective because K_Z is pseudo-effective and $(1 - \delta)D_Z$ is effective. Run its minimal model program. We show inductively that every step is crepant for M , preserves nefness of M , and preserves the Cartier property of ℓM .

At a given step let R be a negative extremal ray for the driving adjoint. Since M is nef and $\delta < 1/2$, the ray is also negative for the klt adjoint $M - D_Z/2$. The length bound gives a rational curve C spanning R such that

$$0 < -(M - D_Z/2) \cdot C \leq 2 \dim Z = 8.$$

Set $a_C = M \cdot C$ and $b_C = D_Z \cdot C$. Then

$$a_C \geq 0, \quad b_C \leq 2a_C + 16, \quad a_C < \delta b_C.$$

It follows that

$$0 \leq a_C < \frac{16\delta}{1 - 2\delta} < \frac{1}{\ell}.$$

Because ℓM is Cartier, $a_C \in \ell^{-1}\mathbb{Z}$, and therefore $a_C = 0$.

The line-bundle clause of the contraction theorem now descends $\mathcal{O}_Z(\ell M)$ to an actual line bundle on the contraction base [12, Theorem 1.1(4)(iii)]; the same theorem's part (5) gives the length bound just used. For a divisorial contraction its pullback is the original bundle. For a flip it pulls back on both sides. Consequently M remains nef, the same multiple ℓM remains Cartier, and its pullbacks to a common resolution agree. This is crepancy for the full adjoint, rather than numerical triviality alone. More explicitly, choose a rational section of the descended line bundle. The divisor ℓM differs from its pullback divisor by a principal divisor; transform the same rational function across the flip. The two exact divisor pullbacks then agree on a common resolution, with compatible canonical divisors. The driving pair stays klt. The boundary with coefficient $1/2$ is smaller than the driving boundary, so it is again klt at the next step. The preceding argument therefore repeats with the same ℓ and δ .

Let Z' be the resulting model, and denote strict transforms by D' , M' , and G' . Put $K' = K_{Z'}$. We have

$$\begin{aligned} (Z', D') \text{ is lc,} \quad K' \text{ is pseudo-effective,} \quad M' \sim_{\mathbb{Q}} G', \quad \kappa(M') \leq 0, \\ M' - uD' \text{ is nef and } (Z', (1 - u)D') \text{ is klt} \quad (0 < u \leq \delta). \end{aligned} \quad (5.5)$$

Here the last nefness assertion follows by interpolation between M' and $M' - \delta D'$. The klt assertion follows by interpolation between the klt driving pair and the lc full-boundary pair. The underlying variety Z' is klt.

We have reduced the problem to a fixed nef interval of actual klt adjoints and a signed representative supported on its boundary. We next use that representative to locate a nonempty boundary floor.

A nonempty floor and the reduced non-klt ideal

Some coefficient of G' is positive. Suppose otherwise. Intersect $K' + D' \sim_{\mathbb{Q}} G' \leq 0$ with the third power of an ample divisor. Pseudo-effectivity of K' and effectivity of D' show that all inequalities are equalities; positivity for nonzero effective divisors then gives $D' = 0$ and $G' = 0$.

To see why this is impossible, take a common smooth resolution V of W and Z' . The pullback to V of the signed divisor J is supported over D_W . Since the programs extract no divisors and $D' = 0$, this pullback is exceptional over Z' . It is rationally linearly equivalent to the pullback of K_Y , and hence is nef. The negativity lemma makes this exceptional divisor nonpositive. Its intersection with an ample divisor to the third power is nonnegative by nefness and nonpositive by its sign; it must vanish. Thus the pullback of K_Y is rationally linearly trivial, contrary to $\kappa(Y, K_Y) = -\infty$.

Write $D' = \sum_j D'_j$ and $G' = \sum_j g_j D'_j$, allowing $g_j = 0$. Let $a = \max_j g_j > 0$. Choose a sufficiently small positive rational s and define

$$B = (1 - s)D' + \frac{s}{a}G', \quad \alpha = 1 + \frac{s}{a}, \quad N = K' + B. \quad (5.6)$$

Choose s so that every coefficient $1 - s + (s/a)g_j$ is nonnegative and $s/\alpha \in (0, \delta)$. All these coefficients are at most one, with equality precisely when $g_j = a$. Therefore

$$0 \leq B \leq D', \quad S := \lfloor B \rfloor \neq 0, \quad N \sim_{\mathbb{Q}} \alpha M' - sD'.$$

The divisor N is nef by Equation (5.5). The pair (Z', B) is lc and klt outside S . Indeed, outside S its finitely many boundary coefficients are bounded above by a number less than one; interpolate the lc full-boundary pair with the underlying klt variety. The same reasoning shows that (Z', bB) is klt whenever $0 < b < 1$.

The multiplier ideals are actual ideals

$$\mathcal{J}(Z', B) = \mathcal{I}_S, \quad \mathcal{J}(Z', bB) = \mathcal{O}_{Z'} \quad (0 < b < 1). \quad (5.7)$$

Here S carries its reduced induced structure. To verify the first identity, use the discrepancy description on a log resolution. Since the pair is lc, a regular function satisfies every positive log-discrepancy condition automatically. At a log-discrepancy-zero divisor it must vanish to positive order. All such centers lie in S , and each component of S itself gives such a condition. A function vanishing on reduced S has positive order at every valuation centered in S ; conversely the component valuations force vanishing on S . This proves the identity. These are multiplier ideals for an lc pair, not a substitution of a non-lc ideal for the reduced ideal.

Lifting sections from the floor

We claim that, for every sufficiently large and divisible integer m , the following restriction map to the whole reduced floor S is surjective:

$$H^0(Z', \mathcal{O}_{Z'}(mN)) \longrightarrow H^0(S, \mathcal{O}_S(mN)). \quad (5.8)$$

The interior injectivity theorem will compare the boundary B with $b_0 B$ for $b_0 < 1$, where the pair is klt. It also requires an endpoint $b_2 B$ beyond B . The corresponding residual bundle is $(m - 1)N - (b_2 - 1)B$; its positivity is supplied by the interval Equation (5.5). We now verify this for both endpoints.

Fix rational numbers $0 < b_0 < 1 = b_1 < b_2$ and a projective log resolution $h : R \rightarrow Z'$ of (Z', D') , hence also of (Z', B) . Choose a fixed effective integral h -exceptional divisor A_R , with coefficients large enough that

$$C_i = A_R + h^*(K' + b_i B) - K_R \geq 0 \quad (i = 0, 1, 2).$$

Their supports lie in one simple normal crossings divisor, and $C_0 \leq C_1 \leq C_2$. Moreover C_1 is the strict interior convex combination of C_0 and C_2 with parameter $(1 - b_0)/(b_2 - b_0)$. For mN Cartier put

$$L_R = A_R + h^*(mN) - K_R.$$

This is an integral divisor. Its endpoint residuals are

$$\begin{aligned} L_R - C_i &= h^*((m-1)N + (1-b_i)B) \\ &\sim_{\mathbb{Q}} h^*(t_i(m)M' - v_i(m)D'), \\ t_i(m) &= (m-1)\alpha + (1-b_i)s/a, \quad v_i(m) = (m-1)s - (1-b_i)(1-s). \end{aligned} \tag{5.9}$$

For $i = 0, 2$, $t_i(m) > 0$ once m is large, and

$$u_i(m) := \frac{v_i(m)}{t_i(m)} \longrightarrow \frac{s}{\alpha} \in (0, \delta).$$

Consequently each endpoint residual is rationally linearly equivalent to a positive rational multiple of $h^*(K' + (1 - u_i(m))D')$, a nef klt adjoint pulled back to its log resolution. By Theorem 4.1, it admits a semipositive singular metric with zero Lelong numbers at every point. Taking positive rational powers and transporting by the actual rational line-bundle isomorphisms preserves this property. The endpoint hypotheses of Theorem 4.2 are therefore satisfied.

That theorem gives injectivity from the first ordinary cohomology group of $K_R + L_R - [C_1]$ to that of $K_R + L_R - [C_0]$. The precise floor calculation is

$$K_R + L_R - [C_i] = h^*(mN) + [K_R - h^*(K' + b_i B)]. \tag{5.10}$$

Multiplier-ideal local vanishing and the projection formula [12, Theorem 3.3] identify these groups with

$$H^1(Z', \mathcal{O}_{Z'}(mN) \otimes \mathcal{J}(Z', b_i B)).$$

The comparison map is the map induced by inclusion of the multiplier ideals. By Equation (5.7), we have proved injectivity of

$$H^1(Z', \mathcal{O}_{Z'}(mN) \otimes \mathcal{I}_S) \longrightarrow H^1(Z', \mathcal{O}_{Z'}(mN)).$$

The exact sequence of S proves Equation (5.8).

The restriction map is now surjective. To obtain a nonzero global section, we must produce one on the whole reduced scheme S . We do this on a dlt floor using adjunction and threefold abundance. The next step therefore includes both gluing on that floor and actual line-bundle descent to S .

Whole-floor abundance and descent

Take a projective $\mathbb{Q}$ -factorial dlt blowup

$$g : (Q, B_Q) \longrightarrow (Z', B), \quad K_Q + B_Q = g^*(K' + B) = g^*N,$$

extracting only lc places; this is the lc case of the dlt blowup theorem [14, Theorem 2.10]. Set $T = \lfloor B_Q \rfloor$. It is nonempty. Whole-floor dlt adjunction equips T with an effective rational different Diff_T such that

$$K_T + \text{Diff}_T = (K_Q + B_Q)|_T = g^*N|_T \quad (5.11)$$

as the actual rational adjoint line bundle, including the conductor and residue identifications on the normalization. The pair on T is semi-dlt; see [10, Remark 1.2(3)] and [13, Remark 2.7].

We record the depth condition needed for this whole-floor statement. The underlying Q is klt and $\mathbb{Q}$ -factorial. Both $\mathcal{O}_Q$ and $\mathcal{O}_Q(-T)$ are Cohen–Macaulay. For the latter assertion, locally trivialize a Cartier multiple of T and take the normal cyclic index cover. It is finite and quasi-étale, hence klt and Cohen–Macaulay. Its finite pushforward decomposes into the corresponding rank-one reflexive sheaves, including $\mathcal{O}_Q(-T)$; in characteristic zero these are direct summands. They are therefore Cohen–Macaulay. The sequence

$$0 \longrightarrow \mathcal{O}_Q(-T) \longrightarrow \mathcal{O}_Q \longrightarrow \mathcal{O}_T \longrightarrow 0$$

then gives S_2 for T (in fact the required Cohen–Macaulay depth along its support). Its codimension-one crossings and the conductor description come from the dlt structure. Thus the semi-log-canonical adjunction in Equation (5.11) is on the whole projective threefold, not on a disjoint collection of its components.

The adjoint in Equation (5.11) is nef. Projective semi-dlt threefold abundance makes it semiample [10, Corollary 4.10]. Hence

$$H^0(T, \mathcal{O}_T(mg^*N)) \neq 0$$

for all sufficiently divisible positive m .

These sections descend to S . Indeed $\mathcal{J}(Q, B_Q) = \mathcal{O}_Q(-T)$, and crepancy together with multiplier-ideal local vanishing gives

$$g_*\mathcal{O}_Q(-T) = \mathcal{I}_S, \quad R^i g_*\mathcal{O}_Q(-T) = 0 \quad (i > 0). \quad (5.12)$$

One can compute both assertions on a common log resolution of (Q, B_Q) and (Z', B) . The equality of their pulled-back adjoints identifies the discrepancy round-up sheaves, and local vanishing for each log resolution followed by Leray gives Equation (5.12). Since Z' is normal, $g_*\mathcal{O}_Q = \mathcal{O}_{Z'}$. Pushing forward the divisor sequence therefore gives

$$g_*\mathcal{O}_T = \mathcal{O}_S.$$

The projection formula identifies the nonzero section spaces above with $H^0(S, \mathcal{O}_S(mN))$. Choose m both sufficiently divisible for this semiampleness and sufficiently large for Equation (5.8). A nonzero section on T then descends to S and lifts to Z' . We have proved $\kappa(N) \geq 0$.

The final klt adjoint

Because $B \leq D'$, section inclusion in divisible degrees gives $\kappa(N) \leq \kappa(M') \leq 0$. Thus $\kappa(N) = 0$. Now consider the klt pair

$$\left(Z', \left(1 - \frac{s}{\alpha} \right) D' \right).$$

Its nef adjoint is rationally linearly equivalent to N/α , so its Kodaira dimension is zero. By Theorem 4.1, on a smooth projective log resolution a positive Cartier multiple of its

pullback has a semipositive singular metric with zero point Lelong numbers. All hypotheses of the fourfold semiample result of Gongyo–Matsumura are now satisfied: the pair is projective klt with rational boundary, the adjoint has nonnegative Kodaira dimension, and the specified pullback metric exists [17, Corollary 5.3].

Semiample and Kodaira dimension zero make this adjoint numerically trivial. On the other hand, K' is pseudo-effective, $D' \neq 0$ because G' has a positive coefficient, and $1 - s/\alpha > 0$. Intersecting $K' + (1 - s/\alpha)D'$ with the third power of an ample divisor gives a strictly positive number. This contradiction proves the proposition. $\square$

6 Moving jets on a two-slot bundle

We continue with the terminal fourfold Y , its nef non-big canonical divisor K , and the polarizations of Equation (3.1). Thus $n = 4$, the integer $\iota > 0$ makes ιK Cartier, and

$$L = L_t = r_t(K + tA), \quad L^n \longrightarrow \epsilon^n, \quad r_t \longrightarrow \infty.$$

The positive number ϵ is fixed throughout this section. Retain the fixed locus Y^{vg} from Section 3, on which Proposition 2.2 holds and every integral curve has K -degree at least $1/\iota$. In particular,

$$L \cdot C \geq r_t/\iota \quad \text{if } C \text{ passes through a point of } Y^{\text{vg}}. \quad (6.1)$$

A very general locus here means the complement of a countable union of proper closed subsets. We may enlarge that union whenever finitely or countably many additional conditions are imposed.

The scalar estimate of Proposition 3.5 limits vanishing on one copy of Y . We now seek the complementary estimate: a polarization on a projective bundle over $Y \times Y$ generates many jets at a very general point. Its Seshadri constant measures this local positivity [29, Chapter 5].

Failure of the estimate would produce a moving base component. Positive-dimensional fibers over either copy of Y are excluded on a bundle slice, using general type and, for a multisection, Proposition 5.1. If both projections are generically finite, the required contradiction has a different source: ramification bounds reduce the family to finitely many fixed covers, and their birational automorphisms cannot sweep the product. We isolate that argument before applying the moving-center method.

6.1 The bundle and its volumes

Put $S_0 = \iota K$, an actual Cartier divisor on Y . On $Y \times Y$, a subscript denotes pullback from the corresponding factor. Let

$$\pi : Z = \mathbb{P}(\mathcal{O}(S_{0,1}) \oplus \mathcal{O}(S_{0,2})) \longrightarrow Y \times Y, \quad \xi = c_1(\mathcal{O}_Z(1)). \quad (6.2)$$

We use the convention

$$\pi_* \mathcal{O}_Z(a\xi) = \text{Sym}^a(\mathcal{O}(S_{0,1}) \oplus \mathcal{O}(S_{0,2})) \quad (a \geq 0).$$

We suppress pullback symbols for divisors on the base. For a positive integer q , define

$$P = L_1 + L_2 + q\xi, \quad d = \dim Z = 2n + 1 = 9. \quad (6.3)$$

This q is a bundle weight; it is unrelated to the irregularity $q(Y) = 0$.

The two summands determine two disjoint sections, called the axes. The complement of the axes is the torus open of Z . Fixing either base coordinate at a point and trivializing the constant line gives a slice

$$\pi_{\text{sl}} : Z_{\text{sl}} = \mathbb{P}(\mathcal{O}_Y \oplus \mathcal{O}_Y(S_0)) \longrightarrow Y, \quad P_{\text{sl}} = L + q\xi_{\text{sl}}. \quad (6.4)$$

The restriction of P is identified with P_{sl} ; the remaining constant factor is trivialized when restricting sections.

Lemma 6.1. *The varieties $Y \times Y$, Z , and Z_{sl} are projective, $\mathbb{Q}$ -factorial, and klt. The tautological classes of the two bundles are nef, and P and P_{sl} are ample rational divisors. Writing $K_{\Sigma} = K_1 + K_2$ on the product and $K_{\Sigma} = K$ on a slice, one has*

$$K_{\text{bundle}} = -2\xi + (\iota + 1)K_{\Sigma} \leq \frac{\iota + 1}{r_t} P_{\text{bundle}} \quad (6.5)$$

in nef order.

Proof. Let $p : \tilde{Y} \rightarrow Y$ be a smooth projective resolution. Rational singularities and the irregularity conclusion in Proposition 2.1 give $q(\tilde{Y}) = 0$. The product description of the Picard group, or the seesaw principle and the vanishing of $\text{Hom}(\text{Alb}(\tilde{Y}), \text{Pic}^0(\tilde{Y}))$, gives

$$\text{Pic}(\tilde{Y} \times \tilde{Y}) = \text{pr}_1^* \text{Pic}(\tilde{Y}) \oplus \text{pr}_2^* \text{Pic}(\tilde{Y}).$$

For a prime Weil divisor on $Y \times Y$, take its strict transform on this smooth product. Its Cartier class is a sum of pullbacks from the two factors. Pushing down the corresponding linear equivalence expresses the original Weil divisor class as a sum of pullbacks of Weil divisor classes on Y . These are $\mathbb{Q}$ -Cartier because Y is $\mathbb{Q}$ -factorial. Thus $Y \times Y$ is $\mathbb{Q}$ -factorial. Exceptional components have images of codimension at least two and do not affect this divisor-class calculation.

The bundle over $\tilde{Y} \times \tilde{Y}$ is a smooth resolution of Z . Its Picard group is the Picard group of the base plus the integral tautological class. Pushing down as above proves $\mathbb{Q}$ -factoriality of Z . The identical argument over $\tilde{Y}$ proves it for the slice. A product of canonical varieties is canonical, as follows directly from the product of resolutions and the canonical discrepancy formula. Projective bundles are smooth over their bases, so the bundles are klt as well.

The direct sum of nef line bundles is nef; hence its tautological class is nef. That class is also relatively ample. Adding a positive pullback of an ample class therefore makes it ample: for example, write the sum as a positive multiple of a relatively ample class made ample by a sufficiently large base twist, plus nef classes. This proves the assertions about P and P_{sl} . Finally, the canonical bundle formula for a rank-two projective bundle gives the equality in Equation (6.5). Since ξ is nef and $K_{\Sigma} \leq L_{\Sigma}/r_t$, the stated nef inequality follows. $\square$

Lemma 6.2. *Assume $q\iota/r_t \leq 1$. For sufficiently small t , there are positive constants c_n, C_n , depending only on n , such that*

$$c_n q \epsilon^{2n} \leq P^d \leq C_n q \epsilon^{2n}. \quad (6.6)$$

On either slice,

$$0 < P_{\text{sl}}^{n+1} \leq 2^{n+2} q \epsilon^n. \quad (6.7)$$

Proof. For a rank-two split bundle with first Chern classes E_1, E_2 , the projective-bundle formula is

$$\pi_*(\xi^i) = \sum_{a+b=i-1} E_1^a E_2^b \quad (i \geq 1), \quad \pi_*(1) = 0.$$

All classes in the expansion of P^d are nef. Put $L_\Sigma = L_1 + L_2$. Each $E_j = S_{0,j}$ is bounded above in nef order by $(\iota/r_t)L_\Sigma$. Consequently

$$\begin{aligned} dqL_\Sigma^{2n} &\leq P^d \\ &\leq qL_\Sigma^{2n} \sum_{i=1}^d i \binom{d}{i} (q\iota/r_t)^{i-1} \leq d2^{d-1} qL_\Sigma^{2n}. \end{aligned}$$

Here $L_\Sigma^{2n} = \binom{2n}{n}(L^n)^2$, and $L^n \rightarrow \epsilon^n > 0$. This proves Equation (6.6).

On a slice one summand is trivial, so

$$P_{\text{sl}}^{n+1} = \sum_{i=1}^{n+1} \binom{n+1}{i} q^i L^{n+1-i} S_0^{i-1} \leq qL^n \sum_{i=1}^{n+1} \binom{n+1}{i}.$$

For small t , we may use $L^n \leq 2\epsilon^n$, giving Equation (6.7). Positivity follows either from ampleness or from the $i = 1$ term. $\square$

We now choose the parameters in the order required by the proof. After fixing ϵ , choose $\delta > 0$ such that

$$(2/\delta)^n 2^{n+2} \epsilon^n < 1. \quad (6.8)$$

Next choose an integer $q > 0$ so large that the lower bound in Equation (6.6) gives

$$P^d > (2n + 3 + d\delta)^d \quad (6.9)$$

for all sufficiently small t . This is possible because that lower bound is linear in q . The integer q is now fixed. Finally decrease t , imposing $q\iota/r_t \leq 1$ and all the preceding estimates. Further decreases of t do not change ϵ, δ , or q .

6.2 The large Seshadri estimate

For an ample rational divisor P and a smooth point z , write

$$\varepsilon(P; z) = \inf_{C \ni z} \frac{P \cdot C}{\text{mult}_z C},$$

where the infimum is over integral projective curves through z .

Proposition 6.3 (Two-slot jets). *Let Z and P be defined by Equations (6.2) to (6.3). Fix $\epsilon > 0$, choose δ satisfying Equation (6.8), and then fix q large enough for Equation (6.9). For all sufficiently small positive rational t , one has*

$$\varepsilon(P; z) > 2n + 2 = 10 \quad (6.10)$$

at a very general smooth point of the torus open of Z .

6.3 Dominant finite correspondences

The final case of the jet argument will be a family of n -dimensional subvarieties whose two projections to Y are generically finite. The following proposition excludes such a family using only bounds for its degree and canonical intersection. The family itself may change with t .

Proposition 6.4 (Bounded correspondences cannot sweep the bundle). *Fix $\epsilon > 0$, an integer $q > 0$, and positive constants B_0, B_1 . Keep Y, K, A, L_t and $P = L_1 + L_2 + q\xi$ as above. For all sufficiently small positive rational t , there is no irreducible finite-type parameter space T and integral closed subvariety $\Gamma \subset T \times Z$, dominant over T , with the following properties:*

- (i) *The evaluation $\Gamma \rightarrow Z$ is dominant, and a general fiber V over T is an integral subvariety of dimension n .*
- (ii) *On a smooth projective resolution $V^* \rightarrow V$, both projections $g_i : V^* \rightarrow Y$, $i = 1, 2$, are dominant and generically finite.*
- (iii) *Writing P also for its pullback to V^* , one has*

$$0 < P^n \cdot V \leq B_0, \quad K_{V^*} P^{n-1} \leq B_1(P^n \cdot V).$$

The restriction on t depends only on the fixed geometry and ϵ, q, B_0, B_1 , not on the parameter space or family.

Proof. We first rule out branch divisors that sweep Y , using the intersection bounds uniformly in t . We then fix t and the family: a common branch complement gives finitely many covers, whose birational graph families lead to an abelian variety dominating Y .

Since $P - g_i^* L$ is nef, the projection formula gives

$$\deg(g_i) L^n \leq P^n \cdot V \leq B_0. \quad (6.11)$$

The degrees are uniformly bounded because $L^n \rightarrow \epsilon^n > 0$. This degree bound alone does not give a finite list of covers; we must first control their branch loci.

Numerical bounds for branch divisors. Resolve the general members of the incidence family in family. After shrinking the irreducible parameter space, there is a smooth projective family $\mathcal{V}^* \rightarrow T$ with integral fibers birational to the corresponding V , together with the two generically finite morphisms $g_i : V^* \rightarrow Y$ on each fiber. For either projection, let $T_i \rightarrow Y$ denote the finite normal Stein factor. Its function field is $\mathbb{C}(V)$.

If J is a branch prime of $T_i \rightarrow Y$, a ramification prime on T_i has a strict transform R on V^* . A proper birational morphism to a normal variety is an isomorphism at the generic point of each target divisor, so this transform is present. Over the smooth locus of Y , its ramification coefficient is at least one. There is an effective canonical comparison

$$K_{V^*} \sim_{\mathbb{Q}} g_i^* K + E_i, \quad E_i \geq R. \quad (6.12)$$

For completeness, resolve the rational lift to a resolution of the terminal target. Ramification between smooth varieties and the effective terminal discrepancies give this formula upstairs; pushing down to V^* gives Equation (6.12).

Set $d_J = L^{n-1} \cdot J$. Projection and nef order imply

$$0 < d_J \leq P^{n-1} \cdot R \leq K_{V^*} P^{n-1} \leq C_3, \quad (6.13)$$

where $C_3 = B_1 B_0$ is independent of t and of the family. The same estimate applies to every prime divisorial image of ramification that meets Y_{sm} .

We also need a canonical bound on J , which need not be normal. On Y_{sm} , the divisor J is a Gorenstein hypersurface. If $\nu : J^\nu \rightarrow J$ is its normalization, finite duality gives, in codimension one,

$$K_{J^\nu} + D_{\text{cond}} \sim_{\mathbb{Q}} \nu^*((K + J)|_J), \quad D_{\text{cond}} \geq 0.$$

The conductor therefore has the sign that decreases the normalized canonical intersection. Terminal singularities are smooth in codimension two, so $J \cap Y_{\text{sing}}$ has dimension at most $n - 3$. Its inverse image on the normalization has codimension at least two. It introduces no omitted codimension-one term in this comparison. On a resolution J^* , the exceptional canonical cycles push to zero against $n - 2$ pullbacks of L . Consequently

$$\begin{aligned} K_{J^*} L^{n-2} &\leq (K + J) J L^{n-2} \\ &\leq d_J / r_t + d_J^2 / L^n \leq C_4 d_J. \end{aligned} \tag{6.14}$$

Here $K \leq L / r_t$. The square term is the Hodge-index inequality

$$J^2 L^{n-2} \leq \frac{(J L^{n-1})^2}{L^n}.$$

It applies to the $\mathbb{Q}$ -Cartier divisor J ; one can pull back to a resolution and approximate the nef class L by ample classes. The bound in Equation (6.13) and the positive lower bound for L^n make C_4 uniform.

If a family of these prime images dominates Y , choose a very general member and then a very general point on its resolution mapping to Y^{vg} . Such a choice is legitimate despite the countable exceptional set. For each excluded proper closed subset of Y , dominance says that the generic image divisor is not contained in it; omit the corresponding proper parameter locus. On the remaining very general member, omit the inverse images of the countably many excluded subsets, together with the finite birational-system and quasi-finiteness exceptions. The chosen image divisor is of general type by Proposition 2.2. Its top L -degree is d_J , and Equation (6.14) is the required canonical bound. Lemma 3.4 excludes this sweeping family for sufficiently small t .

A fixed branch complement and finitely many covers. The preceding exclusion holds uniformly over all families in the statement. Now fix one such sufficiently small t and its incidence family. No further decrease of t will be made in this proof. The smooth projective family $\mathcal{V}^* \rightarrow T$ has integral fibers, and both projections to the normal variety Y are dominant and generically finite. Their degrees are bounded by Equation (6.11). After a finite parameter extension and shrinking, follow the finitely many geometric generic components of the relative Jacobian over Y_{sm} . Every followed component whose fiber image is a divisor has proper total image closure: the preceding argument excludes a dominant such family using Equations (6.13) and (6.14).

These are exactly the hypotheses of the simultaneous finite-cover lemma [38, Lemma 7.5]. It gives one smooth nonempty open $Y^\circ \subset Y$ over which every general finite Stein cover in either slot is finite étale, and finite lists of the resulting normal covers of Y . The lemma accounts for every branch prime by its ramification strict transform, then applies purity and finiteness of bounded-degree étale covers of the fixed open. In particular, this is a finite-type family argument; a numerical branch bound alone would not suffice. The open and lists may depend on the fixed t and incidence family. No uniformity in those choices is needed.

Countably many birational graph families. Each V gives a birational map between one cover from each list: both finite Stein factors have function field $\mathbb{C}(V)$. Its image in $Y \times Y$ is the finite image of the closure of that birational graph. Fix one pair of covers that are birational. A smooth projective resolution of their common birational class has pseudo-effective canonical class, by the effective canonical comparison with the nef K on Y . It is non-uniruled by [5].

The birational-group structure theorem of Hanamura [19], in the form stated in [4, Proposition 3.7 and Theorem 3.8], permits a smooth projective model U of this class for which the reduced flat-graph birational scheme is a group scheme locally of finite type and

$$A_U = \text{Aut}^0(U)$$

is its identity component, an abelian variety. The flat graphs are represented in graph Hilbert schemes. There are countably many Hilbert polynomials and finitely many components in each finite-type Hilbert scheme, so the birational maps occur in countably many translates $A_U b$.

For such a translate, its graphs are parametrized on a dense open by the rational evaluation map

$$U \times A_U \dashrightarrow U \times U, \quad (x, a) \mapsto (x, a \cdot b(x)). \quad (6.15)$$

If this map is not dominant, its image closure is a proper closed subset of $U \times U$. Fix birational identifications of U with the two covers. Every graph meets the product of their fixed birational opens densely, because it dominates both factors. Transporting the image closure through those opens, then taking its closure in the product of covers, preserves its dimension. The subsequent finite map to $Y \times Y$ also preserves dimension. Thus all correspondences belonging to this translate are contained in a proper closed subset of $Y \times Y$. The possible boundary of an individual birational map introduces no new component: the generic graph has already been included.

There are finitely many cover pairs and countably many translates for each pair. If every map in Equation (6.15) were nondominant, the image of the original incidence would be contained in a countable union of proper closed subsets of $Y \times Y$. But $\Gamma \rightarrow Z$ is dominant, so its composite to $Y \times Y$ is dominant and its image contains a nonempty open subset. Over the uncountable field $\mathbb{C}$, such an open cannot be covered by countably many proper closed subsets. Therefore the map in Equation (6.15) is dominant for at least one translate.

The abelian curve contradiction. For a general x in this last evaluation, $A_U \cdot b(x)$ is a dense orbit in U . A stabilizer element fixes every point of that orbit by commutativity; it fixes U by density. The action of $\text{Aut}^0(U)$ is faithful, so the stabilizer is zero-dimensional. Hence the orbit map from A_U is generically finite dominant. Composing with the rational map through either fixed cover gives a generically finite dominant rational map

$$A_U \dashrightarrow Y.$$

In particular $\dim A_U = \dim Y = n$.

Resolve this map as $\pi_A : T_A \rightarrow A_U$ and $g : T_A \rightarrow Y$, with T_A smooth and projective. Since the canonical bundle of A_U is trivial, and since Y is terminal, the canonical comparisons give

$$K_{T_A} \sim_{\mathbb{Q}} E_{\pi_A}, \quad K_{T_A} \sim_{\mathbb{Q}} g^* K + R, \quad E_{\pi_A}, R \geq 0, \quad (6.16)$$

where E_{π_A} is exceptional over A_U . The images of the exceptional divisors of π_A have codimension at least two on the smooth abelian variety.

Choose a point in the isomorphism open of π_A where g is quasi-finite, outside $\text{Supp}(R)$, and mapping to Y^{vg} . Dominance and the countable-exclusion description ensure that such a point exists. A general sufficiently ample complete-intersection curve in A_U through its image avoids all the exceptional centers: they have codimension at least two and do not contain the prescribed point. Its strict transform C is disjoint from E_{π_A} and is not contained in R . The map g is nonconstant on C , since it is quasi-finite at the chosen point. Intersecting Equation (6.16) with C gives

$$0 = K_{T_A} \cdot C = g^* K \cdot C + R \cdot C > 0.$$

The final strict inequality uses $K \cdot g(C) \geq 1/\iota$, the defining curve property of Y^{vg} , multiplied by the positive degree of $C \rightarrow g(C)$, together with $R \cdot C \geq 0$. This contradiction excludes the remaining correspondence family. $\square$

6.4 Excluding moving base components

Proof of Proposition 6.3. All constants used to exclude moving centers below are independent of the section degree and of small t , once ϵ, δ, q have been fixed. We first produce such a center from failure of Equation (6.10); we then exclude each possible dimension of its fibers over the two factors.

A moving component from failure of the estimate. Suppose that Equation (6.10) fails at very general torus points. At such a point there is a curve whose ratio is strictly less than $2n + 3$. This conclusion does not require the infimum defining the Seshadri constant to be attained.

Take $d + 1$ levels

$$\tau_j = 2n + 3 + j\delta, \quad 0 \leq j \leq d.$$

By Equation (6.9), ample Hilbert asymptotics and the dimension of the jet space at a smooth point show that, for large divisible k , every kernel of jets of order $\lceil k\tau_j \rceil$ is nonzero. Explicitly, the leading coefficients of the two dimensions are $P^d/d!$ and $\tau_j^d/d!$. If a section in the lowest kernel does not contain the curve just chosen, its local intersection with that curve at the marked point is at least $\lceil k\tau_0 \rceil \text{mult}_z C$. This exceeds $kP \cdot C$. Thus every section in that kernel contains the curve, and the lowest base locus has a positive-dimensional component through the mark.

These conditions produce a dominant marked-point family in the sense of Lemma 3.3. For generality, choose the marked point outside the proper rank-jumping and base-dimension loci for all divisible degrees; there are only countably many such conditions. For the degree k now chosen, restrict to the open where the finitely many jet kernels have constant rank. The dimension of the base scheme at its marked point is upper semicontinuous. The existence of the witnessing curve at very general marks therefore forces the positive-dimensional-base condition on a dominant locus; no family of the witnessing curves is assumed. The finitely many generic base components and their inclusions can then be followed after a finite parameter extension and shrinking.

We may increase the divisible integer k to arrange, in addition, that all divisors

$$kL + (2kq - j)S_0, \quad 0 \leq j \leq kq, \tag{6.17}$$

are Cartier and globally generated. Here is one simultaneous verification. For $1 \leq a \leq n$, subtract aA and then K . The resulting class is

$$(kr_t + (2kq - j)\iota - 1)K + (kr_t - a)A,$$

which is ample once $kr_t t > n$ and $kr_t \geq 1$. Klt Kawamata–Viehweg vanishing followed by regularity with respect to A proves global generation for every j in Equation (6.17).

The moving-multiplicity lemma supplies an irreducible parameter space T , a dominant incidence $\Gamma \subset T \times Z$, and integral fiber components $V = V_u$ of a fixed dimension $1 \leq f < d$. The high-kernel subseries has V as an isolated base component generically, with ordinary generic order at least

$$h = \lceil k\delta/2 \rceil, \quad k/h \leq 2/\delta.$$

For a smooth resolution V^* , the base-component estimates and Equation (6.5) give

$$0 < P^f \cdot V \leq (2/\delta)^{d-f} P^d, \quad (6.18)$$

$$K_{V^*} P^{f-1} \leq \left(\frac{\iota + 1}{r_t} + \frac{2(d-f)}{\delta} \right) P^f \cdot V. \quad (6.19)$$

The right sides are bounded uniformly in the specified sense.

We first exclude positive-dimensional slot fibers. The chosen incidence point will ensure that all image subvarieties below meet the prescribed very general locus of Y . For an intermediate fiber F in a slice, let $W \subseteq Y$ be its image and write $s = \dim F$, $w = \dim W$. The three possibilities are summarized below; the subsequent paragraphs prove all the indicated exclusions.

Dimensions	Geometry	Contradiction
$s = w < n$	Generically finite over a proper image	General type and the degree obstruction on F .
$s = w + 1$, $w < n$	Entire bundle over W	Direct slice-volume bound if $w = 0$; fillers and subadjunction on W if $w > 0$.
$s = w = n$	Non-axis multisection over Y	Its two axis intersections give a signed representative; the logarithmic degree obstruction applies.

A full slot fiber is excluded by exchanging slots. This leaves generically finite projections. Proper images are again excluded by general type, and dominant images are excluded by Proposition 6.4.

Restriction to a slot fiber. Consider V over either factor of $Y \times Y$. Choose a very general incidence point first. We require that its two base coordinates belong to Y^{vg} , that it lies in the torus, and that the ordinary ideal-power and isolated-component conclusions above hold near it. We also impose the smoothness conditions for the incidence over T and for V over the smooth open of its actual slot image. These are available on dense opens.

Now take the slot value of this point, and let F be the reduced fiber component through it. Write $s = \dim F$, the general fiber dimension. Suppose first that

$$0 < s < n + 1.$$

The restricted high-kernel subseries on the opposite slice is nonzero and has F as an isolated base component generically. Indeed, near the selected point the original base support is just V . No other base component can contain the fiber component through that point. The scheme fiber over the ambient slot agrees with the scheme fiber over the actual image, and smoothness over that image makes it reduced at the selected generic component. Thus the image of $\mathcal{I}_V^h$ in the slice is contained in $\mathcal{I}_F^h$, as an ordinary ideal power. A restriction that vanished identically on the whole slice would contradict this local proper base support.

Applying Lemma 3.2 in the slice gives

$$0 < P_{\text{sl}}^s \cdot F \leq (2/\delta)^{n+1-s} P_{\text{sl}}^{n+1}, \quad (6.20)$$

$$K_{F^*} P_{\text{sl}}^{s-1} \leq \left(\frac{\iota + 1}{r_t} + \frac{2(n+1-s)}{\delta} \right) P_{\text{sl}}^s \cdot F. \quad (6.21)$$

Let $W \subseteq Y$ be the image of F in the other base, and put $w = \dim W$. It meets the prescribed very general locus. Since the bundle fiber has dimension one, either $w = s$ or $w = s - 1$. We treat every possibility.

A generically finite proper image. If $w = s < n$, the resolution of W is of general type by Proposition 2.2. The same holds for F^* : after resolving the generically finite map to a resolution of W , ramification adds an effective divisor to the pulled-back canonical divisor. The polarization P_{sl} dominates the pullback of L . Equations (6.20) and (6.21) therefore contradict Lemma 3.4 for sufficiently small t .

The whole bundle over a proper image. Suppose $s = w + 1$, where necessarily $w < n$. A positive-dimensional closed subset of the generic $\mathbb{P}^1$ fiber is the entire fiber. Consequently

$$F = \pi_{\text{sl}}^{-1}(W)$$

as reduced integral subvarieties. Nef intersection and the projective-bundle formula give

$$qL^w \cdot W \leq P_{\text{sl}}^{w+1} \cdot F. \quad (6.22)$$

For $w = 0$, the left side is q . The right side is at most

$$(2/\delta)^n P_{\text{sl}}^{n+1} \leq (2/\delta)^n 2^{n+2} q \epsilon^n < q$$

by Equation (6.8), a contradiction.

Assume $w > 0$. We will apply the degree obstruction to W , rather than to the ruled variety F . Expand the restricted high sections using the two homogeneous fiber coordinates. Their coefficients of weight j , for $0 \leq j \leq kq$, are actual sections of

$$\mathcal{O}_Y(kL + jS_0).$$

Let $\mathfrak{a}$ be the ideal generated by all these coefficients in the regular local ring

$$R = \mathcal{O}_{Y, \eta_W}, \quad \mathfrak{m} = \mathfrak{m}_R.$$

This ring is regular because the generic point of W lies in the smooth locus of Y .

We need both the order and the support of this coefficient ideal. After choosing frames, the local ring of the slice at the generic point of F is

$$R[u]_{\mathfrak{m}R[u]}.$$

The residue of u is transcendental over the residue field of R . If a coefficient had order less than h , the least $\mathfrak{m}$ -adic homogeneous part of its polynomial would remain nonzero after adjoining that residue variable. Therefore membership of every high section in $\mathfrak{m}^h R[u]_{\mathfrak{m}R[u]}$ implies $\mathfrak{a} \subseteq \mathfrak{m}^h$. Moreover, if a prime $\mathfrak{p} \subsetneq \mathfrak{m}$ contained $\mathfrak{a}$, its extension to this local polynomial ring would be a proper prime strictly below the generic ideal of F containing the whole high base ideal. This contradicts the isolated support of F . Hence

$$\mathfrak{a} \subseteq \mathfrak{m}^h, \quad \sqrt{\mathfrak{a}} = \mathfrak{m}. \quad (6.23)$$

Multiply every weight- j coefficient by the globally generated space in Equation (6.17). The products all belong to one linear subseries of

$$H^0(Y, \mathcal{O}_Y(2k(L + qS_0))).$$

At η_W , each filler system generates a unit after choosing a frame. Thus this new subseries has precisely the joint ideal $\mathfrak{a}$, not merely an ideal with the same support. Its threshold construction in Lemma 3.2 gives an effective rational divisor

$$\Theta \sim_{\mathbb{Q}} c 2(L + qS_0), \quad 0 < c \leq (n - w)k/h,$$

such that the pair is lc at the generic point of W and has an lc place centered there. Apply Lemma 3.1 with the testing class L . Since $qS_0 \leq L$ in nef order, we obtain

$$\begin{aligned} K_{W^*} L^{w-1} &\leq (K + 2c(L + qS_0)) L^{w-1} \cdot W \\ &\leq \left(\frac{1}{r_t} + \frac{8(n-w)}{\delta} \right) L^w \cdot W. \end{aligned} \quad (6.24)$$

The top degree $L^w \cdot W$ is bounded by Equation (6.22) and Equation (6.20). The variety W^* is of general type. Equation (6.24) and the top-degree bound now contradict Lemma 3.4 on W^* .

A non-axis multisection over all of Y . The only remaining intermediate-fiber case is $s = w = n$. Here F is a multisection of positive degree e over Y . It is not an axis, since it contains the selected torus point. Let Σ_0, Σ_∞ be the axes on the slice, with classes

$$\Sigma_0 \sim \xi_{\text{sl}}, \quad \Sigma_\infty \sim \xi_{\text{sl}} - S_0.$$

The intersections with F are effective integral cycles. Their pushforwards define effective integral Weil divisors D_0, D_∞ on Y , allowing a divisor to be zero. Projection and rational equivalence of cycles give

$$D_0 - D_\infty \sim eS_0 = e\iota K. \quad (6.25)$$

This is linear equivalence of Weil divisors: codimension-one rational equivalence on the normal variety Y is precisely linear equivalence. In particular,

$$J = \frac{D_0 - D_\infty}{e\iota} \sim_{\mathbb{Q}} K$$

is an actual signed rational representative.

For either axis, $P_{\text{sl}} - q\Sigma$ is nef: it is L or $L + qS_0$. As $F \cap \Sigma$ is effective and $P_{\text{sl}} \geq \pi_{\text{sl}}^* L$ in nef order,

$$qL^{n-1} \cdot D_i \leq qP_{\text{sl}}^{n-1} \cdot (F \cap \Sigma_i) \leq P_{\text{sl}}^n \cdot F, \quad i = 0, \infty. \quad (6.26)$$

Take a log resolution $p : Y^\dagger \rightarrow Y$ of $D_0 + D_\infty$, and let G be the reduced snc divisor consisting of their strict supports and all exceptional divisors. By Proposition 5.1, $K_{Y^\dagger} + G$ is big. The exceptional terms push to zero against L^{n-1} , while the reduced strict support is bounded by the integral divisor $D_0 + D_\infty$. Hence

$$(K_{Y^\dagger} + G)L^{n-1} \leq \frac{L^n}{r_t} + L^{n-1} \cdot (D_0 + D_\infty). \quad (6.27)$$

The second term is uniformly bounded by Equation (6.26) and Equation (6.20). Also L^n is bounded above and bounded away from zero. Thus Equation (6.27) is the canonical-to-volume bound in the log version of Lemma 3.4, with polarization p^*L and map p . That lemma gives the contradiction in this final intermediate-fiber case.

Reduction to finite correspondences. We have excluded all slot-fiber dimensions $1, \dots, n$. A fiber dimension $n + 1$ over one slot image W would force V to be the entire bundle over $W \times Y$. Since $V \neq Z$, one has $\dim W < n$; its fiber over the other slot then has dimension $\dim W + 1$, which has just been excluded. It follows that both projections of V to Y are generically finite onto their images.

If one image were proper, it would have dimension $f < n$ and be of general type. Its generically finite cover V^* would also be of general type, and the total bounds Equations (6.18) to (6.19) would contradict Lemma 3.4. Therefore

$$f = n, \quad g_i : V^* \longrightarrow Y \quad (i = 1, 2)$$

are generically finite dominant morphisms. The bounds Equations (6.18) to (6.19) supply constants B_0, B_1 independent of small t , of the section degree k , and of the family. We are therefore in the situation excluded by Proposition 6.4.

Every possibility for a moving component has now been excluded. The slice degree obstructions and Proposition 6.4 impose restrictions on t independent of k and of the moving family. The assumed failure of Equation (6.10) is therefore impossible for all sufficiently small t , as required. $\square$

6.5 One finite system of jets on a smooth model

We now convert the strict Seshadri bound into a single surjective jet-evaluation map on a smooth model. Choose t sufficiently small for both Propositions 3.5 and 6.3, and keep it fixed. The next corollary then supplies a point, a Cartier multiple, and finitely many sections whose jets span the required target. These are the finite data used in reduction to positive characteristic; no family of jet systems over varying t is needed.

Corollary 6.5. *Fix sufficiently small t as in Propositions 3.5 and 6.3, and let $p : W \rightarrow Y$ be any fixed smooth projective resolution. Continue to denote pullbacks by L and S_0 . On*

$$Z_W = \mathbb{P}(\mathcal{O}_W(S_0)_1 \oplus \mathcal{O}_W(S_0)_2) \longrightarrow W \times W, \quad M = L_1 + L_2 + q\xi,$$

there are a smooth torus point z_ , over the isomorphism opens of p in both slots, and a sufficiently divisible integer $k_0 > 0$ such that $k_0 L$ is Cartier and*

$$H^0(Z_W, \mathcal{O}_{Z_W}(k_0 M)) \longrightarrow \mathcal{O}_{Z_W}(k_0 M) \otimes (\mathcal{O}_{Z_W, z_*} / \mathfrak{m}_{z_*}^{10k_0+1}) \quad (6.28)$$

is surjective. In particular, finitely many sections can be fixed whose jets span its target.

Proof. Choose a very general smooth torus point z of Z above the isomorphism open of p in both slots and satisfying Equation (6.10), so that $\varepsilon(P; z) > 10$. If $b : \widehat{Z} \rightarrow Z$ is its blowup and E_z is the exceptional divisor, the Seshadri criterion shows that the following rational divisor is ample:

$$b^*P - 10E_z.$$

Indeed one may choose a rational number strictly between 10 and $\varepsilon(P; z)$, and then use the usual ample interval below the Seshadri threshold. Choose $a > 0$ so that aP and $a(b^*P - 10E_z)$ are Cartier. Serre vanishing, applied to the fixed sheaf $\mathcal{O}_{\widehat{Z}}(-E_z)$, gives

$$H^1(\widehat{Z}, \mathcal{O}_{\widehat{Z}}(kb^*P - (10k+1)E_z)) = 0$$

for all sufficiently large multiples k of a . The standard local calculation for the blowup of a smooth point identifies the pushforward of $\mathcal{O}_{\widehat{Z}}(-(10k+1)E_z)$ with $\mathfrak{m}_z^{10k+1}$, with vanishing higher direct images. The restriction sequence therefore gives surjectivity onto jets through order $10k$ at z .

The natural morphism $Z_W \rightarrow Z$ is an isomorphism near the lift z_* of z . Pullback preserves the sections in this jet surjection and identifies their jets. Increase the divisible integer k if necessary so that kL is Cartier, and call it k_0 . Choosing finitely many inverse images of a basis of the finite-dimensional jet target proves the assertion. $\square$

7 Fixed finite data and the final contradiction

The scalar bound and the two-slot jets will now be tested against one independent theorem. We state all its hypotheses before constructing the remaining data. Its independent proof in [38] uses neither abundance nor logarithmic subadditivity.

Theorem 7.1 (Finite-data Frobenius incompatibility). *Let W be a smooth connected projective complex variety of dimension $n \geq 2$. Fix rational Cartier divisors L, H , with H ample, an integral Cartier divisor S , an integer $q > 0$, and real numbers $r > 1$, $\epsilon > 0$. Suppose*

$$H^n \leq (2\epsilon)^n, \quad K_W H^{n-1} \leq 2H^n/r, \quad (2L + qS)H^{n-1} \leq 4H^n, \\ (14\epsilon)^n < 1/4, \quad 80\epsilon < 3/4.$$

The following three kinds of fixed data cannot all exist.

- (i) *A smooth integral complete-intersection flag ending in a curve C , of successive divisor classes $l_i H$ for positive integers l_i , with $\mu_{\min}(\Omega_W^1|_C) \geq 0$.*
- (ii) *On $Z_W = \mathbb{P}(\mathcal{O}(S)_1 \oplus \mathcal{O}(S)_2) \rightarrow W \times W$, in the symmetric-power convention, put $M = L_1 + L_2 + q\xi$. For some integer $k_0 > 0$ with $k_0 L$ Cartier, finitely many sections of $k_0 M$ surject onto $\mathcal{O}_{Z_W, z_*}/\mathfrak{m}_{z_*}^{(2n+2)k_0+1}$ after trivialization at a smooth torus point z_* .*
- (iii) *A point $x \in W$, the blowup $\pi_x : \widehat{W} \rightarrow W$ with exceptional divisor J_x , and a curve class $\gamma = f_*(A_1 \cdots A_{n-1})$, where $f : V \rightarrow \widehat{W}$ is a fixed birational morphism from a smooth integral projective variety and the A_i are ample Cartier divisors, such that $J_x \gamma > 0$ and*

$$\pi_x^*(L + K_W/2 + \lambda S)\gamma \leq 40\epsilon J_x \gamma \quad (0 \leq \lambda \leq q).$$

It suffices to check the two endpoint inequalities.

No nefness of L, S, K_W is assumed, and z_ need not lie over x . All divisors, the flag, the point, the curve representative, and the finite jet sections are fixed over $\mathbb{C}$ before the residual characteristic varies.*

The proof is [38, Theorem 9.1]. We shall apply it with $n = 4$, $S = S_0 = \iota K$, and $r = r_t$. The remaining work is to obtain its ample perturbation, flag, and actual movable curve witness from the fourfold geometry.

7.1 Fixing the polarization, flag, and movable test

Choose a positive rational number ϵ sufficiently small that

$$(14\epsilon)^n < \frac{1}{4}, \quad 80\epsilon < \frac{3}{4}. \quad (7.1)$$

Choose q as in Proposition 6.3, then a sufficiently small rational $t > 0$ so that both jet estimates hold, $r_t > q\iota + 1$, and $L^n < (2\epsilon)^n$. Fix t and r_t . Let $W \rightarrow Y$ be a projective resolution, and continue to denote pullbacks by K, A, L . Terminality gives

$$K_W = K + E, \quad E \geq 0 \text{ exceptional}, \quad S_0 = \iota K \text{ Cartier.}$$

Since L is pulled back from an ample divisor on Y ,

$$K_W L^{n-1} = K L^{n-1} \leq L^n / r_t.$$

Perturb L by a sufficiently small ample rational class on W . By continuity, the resulting ample rational divisor H satisfies

$$H^n \leq (2\epsilon)^n, \quad K_W H^{n-1} \leq \frac{2H^n}{r_t}, \quad (2L + qS_0)H^{n-1} \leq 4H^n. \quad (7.2)$$

For the last inequality, at $H = L$ use $S_0 L^{n-1} \leq \iota L^n / r_t$ and $q_t < r_t$. We now fix H ; all further geometric data will be chosen once over $\mathbb{C}$ before reduction.

The divisor K_W is pseudo-effective. Generic semipositivity, in the smooth empty-boundary case of [6, Theorem 2.1], gives nonnegative H^{n-1} -slope to every positive-rank torsion-free quotient of Ω_W^1 . Apply the Mehta–Ramanathan restriction theorem to the Harder–Narasimhan filtration and its factors [33, Theorem 6.1 and Remark 6.2]. We obtain a fixed flag of smooth integral general very ample sections, of types

$$h_i = l_i H \quad (1 \leq i \leq n-1),$$

ending in a smooth curve C , such that

$$\mu_{\min}(\Omega_W^1|_C) \geq 0. \quad (7.3)$$

The l_i are sufficiently large divisible integers, chosen successively. The final general curve avoids the codimension-two loci where the filtration factors fail to be locally free. Set $\Lambda = \prod_{i=1}^{n-1} l_i$. Then

$$[C] = \Lambda H^{n-1}, \quad h_i \cdot C = l_i \Lambda H^n. \quad (7.4)$$

Next choose a very general point $x \in W$ above the isomorphism locus of $W \rightarrow Y$, and let $\pi_x : \widehat{W} \rightarrow W$ be its blowup, with exceptional divisor J_x . Put

$$D^+ = 2r_t(K + 2tA) + 2E.$$

By Proposition 3.5, every nonzero section of a sufficiently divisible multiple kD^+ satisfies

$$\text{ord}_x(s) \leq 32k\epsilon. \quad (7.5)$$

Indeed, pushing down removes only the effective exceptional part, and is an isomorphism near x .

Lemma 7.2. *There is a strongly movable curve class γ on $\widehat{W}$, represented as the pushforward of an ample complete intersection on a fixed smooth projective birational model, such that*

$$(\pi_x^* D^+) \cdot \gamma < 40\epsilon J_x \cdot \gamma, \quad J_x \cdot \gamma > 0. \quad (7.6)$$

Proof. If $\pi_x^* D^+ - 40\epsilon J_x$ were pseudo-effective, then for any rational c with $32\epsilon < c < 40\epsilon$ the divisor $\pi_x^* D^+ - cJ_x$ would be big: it is a positive convex combination of the preceding pseudo-effective divisor and the big divisor $\pi_x^* D^+$. A sufficiently divisible effective multiple would contradict Equation (7.5). Thus $\pi_x^* D^+ - 40\epsilon J_x$ is not pseudo-effective.

Movable-curve duality [5, Theorem 2.2] gives a strongly movable class with negative pairing against that divisor. Since the inequality is strict, we may take an actual pushforward of an ample complete intersection on one smooth birational projective model, approximating the ample classes rationally and clearing their denominators if needed. The big divisor $\pi_x^* D^+$ has positive pairing with this nonzero class: write it as an ample class plus an effective class and pull back to the chosen model. The strict inequality therefore implies $J_x \cdot \gamma > 0$. $\square$

We omit π_x^* in subsequent intersection formulas involving γ . The concrete complete-intersection representative will continue to test effective divisors after reduction.

7.2 Endpoint comparison

For $0 \leq \lambda \leq q$, put $Q_\lambda = L + K_W/2 + \lambda S_0$. The actual divisor classes on the fixed resolution satisfy

$$D^+ - Q_\lambda = (r_t - 1/2 - \lambda t)K + 3r_t t A + \frac{3}{2}E. \quad (7.7)$$

The first coefficient is positive because $r_t > q t + 1$. The pullbacks of K, A are nef, and E is effective. A strongly movable curve pairs nonnegatively with each of these classes. Consequently Lemma 7.2 gives

$$Q_\lambda \gamma \leq D^+ \gamma < 40\epsilon J_x \gamma \quad (0 \leq \lambda \leq q).$$

This verifies the curve inequalities in Theorem 7.1. It does not require a pseudo-effective cone to specialize to positive characteristic: the curve has the fixed ample complete-intersection representative required by that theorem.

Take z_*, k_0 , and the finite sections supplied by Corollary 6.5 on this resolution. For clarity about the finite choices in the comparison, choose also an integral ample divisor T_0 so that each $T_0 + aS_0$, for $0 \leq a \leq (k_0 - 1)q$, has fixed sections nonvanishing at the two base coordinates of z_* . Serre generation permits these finitely many choices. They supply the residual degrees when the common proof puts $B_p = k_0 \lfloor p/k_0 \rfloor L + T_0$. The differences $B_p - pL$ belong to a fixed finite list.

We have now verified all the hypotheses of Theorem 7.1. The choices were made in the order $n = 4, \epsilon$; then the gap δ and integer q ; then t, r_t ; then the resolution, H , flag, point, and curve; and finally the finite jets and fillers. Only after these choices does its proof let p tend to infinity.

7.3 The two orders of the determinant

We recall the comparison in the proof of [38, Theorem 9.1], identifying the determinant bundles to which our movable test applies. Spread the fixed data as in that proof and work on a smooth reduction in sufficiently large characteristic p . The same notation denotes the reduced divisors and witnesses. For relative Frobenius $F : W \rightarrow W'$, let S'_0 be the base-field twist of S_0 and put

$$\mathcal{E} = F_* \mathcal{O}_W(B_p), \quad s = \text{rk } \mathcal{E} = p^n, \quad R = s^2 = p^8, \quad U_a = H^0(W, \mathcal{O}_W(B_p + paS_0)).$$

The projection formula gives an evaluation map

$$\Phi_p : \bigoplus_{\substack{a+b=q \\ a,b \geq 0}} U_a \otimes U_b \otimes \mathcal{O}_{W' \times W'}(-aS'_{0,1} - bS'_{0,2}) \longrightarrow \mathcal{E} \boxtimes \mathcal{E}. \quad (7.8)$$

Products of the fixed jet sections, multiplied by the fixed fillers, generate the local quotient by the p -th powers of the parameters at z_* . Let A_* be the tensor product of the two base Frobenius-fiber algebras. If the torus coordinate z has value $z_0 \neq 0$ at z_* , this quotient is $A_*[z]/(z^p - z_0^p)$. Project to the coefficient of 1 in its A_* -basis $1, z, \dots, z^{p-1}$. Exactly the weights divisible by p survive, showing that the values of (7.8) span A_* . Its dimension is R , so Φ_p has generic rank R .

On the diagonal, the columns instead restrict to global sections of one line bundle on the Frobenius fiber product:

$$\Delta_F = W \times_{W'} W, \quad \mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pqS_{0,1}).$$

Here $pS_{0,1}$ and $pS_{0,2}$ are isomorphic line bundles on Δ_F , so all the weight pairs $a + b = q$ give this same bundle. The rank at every diagonal point is at most $h^0(\Delta_F, \mathcal{L}_p)$. The diagonal-ideal filtration has graded bundles

$$\mathcal{G}_j = \mathcal{O}_W(2B_p + pqS_0) \otimes A_j(\Omega_W^1), \quad 0 \leq j \leq n(p-1),$$

where A_j is the degree- j part of the symmetric algebra modulo the p -th powers of its local generators. If $e_j = \text{rk } A_j$, then $\sum_j e_j = p^n$. The slope estimate and our one fixed flag give the common bound proved in the companion:

$$h^0(W, \mathcal{G}_j) \leq e_j p^n (7^n H^n + O(p^{-1})).$$

The error is independent of j . Summing therefore bounds every diagonal rank by $R(7^n H^n + O(p^{-1})) < R/4$, since $7^n H^n \leq (14\epsilon)^n < 1/4$.

Choose R generically independent columns of Φ_p , and let m_i be the sum of their weights in slot i . Their nonzero determinant is a section

$$0 \neq \sigma \in H^0(W' \times W', \mathcal{Q}_1 \boxtimes \mathcal{Q}_2), \quad \mathcal{Q}_i = (\det \mathcal{E})^{\otimes s} \otimes \mathcal{O}_{W'}(m_i S'_0).$$

These are actual line bundles. Under the numerical identification with the base-field twist, their first Chern classes satisfy

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{B_p}{p} + \frac{p-1}{2p} K_W + \lambda_i S_0 = Q_{\lambda_i} + \frac{B_p - pL - K_W/2}{p}, \quad \lambda_i = \frac{m_i}{R} \in [0, q]. \quad (7.9)$$

The numerator of the last term belongs to a fixed finite list. Thus the endpoint comparison bounds the intersection of each normalized class with γ by $(40\epsilon + O(p^{-1}))J_x\gamma$, uniformly in the chosen columns.

This intersection bound controls sections on the reduction itself. Put $x' = F(x)$. For any nonzero section of $\mathcal{Q}_i$, the strict transform of its zero divisor on the blowup at x' is effective and pairs nonnegatively with the twist of γ : that curve is still the pushforward of the fixed ample complete intersection. Since $J_x\gamma > 0$, every such section has order at x' at most $R(40\epsilon + O(p^{-1}))$. Bases adapted to vanishing order in the two factors show that a nonzero section of their external product has order at most the sum of these bounds. Within each bidegree the leading tensors are linearly independent, and different bidegrees cannot cancel. Consequently

$$\text{ord}_{(x', x')} \sigma \leq R(80\epsilon + O(p^{-1})).$$

But the rank of Φ_p at (x', x') is less than $R/4$. Constant row operations make more than $3R/4$ rows of the selected matrix vanish at that point, forcing $\text{ord}_{(x', x')} \sigma \geq 3R/4$. The two orders contradict $80\epsilon < 3/4$ for large p . The generic-rank calculation used z_* , whereas this diagonal comparison uses the independently chosen point x .

Thus the terminal counterexample in Proposition 2.1 cannot exist. Canonical section spaces are preserved by its birational construction, so this proves Theorem 1.1.

8 A section on the original normal variety

The geometric argument produces a pluricanonical section on a smooth fourfold. Hashizume's reduction gives an effective real representative of an lc adjoint on the original variety. The following elementary lemma turns that representative into a rational one. It concerns finite linear algebra and uses no nonvanishing or abundance theorem.

Lemma 8.1 (Finite rational feasibility). *Let $M = (m_{ij}) \in \text{Mat}_{w \times v}(\mathbb{Q})$ and $d \in \mathbb{Q}^w$. If the system*

$$d_i - y_i = \sum_{j=1}^v m_{ij} b_j \quad (1 \leq i \leq w), \quad y_i \geq 0 \quad (8.1)$$

has a real solution, it has a rational solution. Moreover, one may require precisely the same y_i to be zero as in the given solution. Consequently, if D is a rational Weil divisor on a normal integral variety X over a field k and $D \sim_{\mathbb{R}} G$ for an effective real Weil divisor G , then $D \sim_{\mathbb{Q}} G_{\mathbb{Q}}$ for an effective rational divisor with $\text{Supp } G_{\mathbb{Q}} = \text{Supp } G$. If rD is Cartier for a prescribed positive integer r , then $H^0(X, \mathcal{O}_X(mrD)) \neq 0$ for some $m > 0$.

Proof. Fix the equations $y_i = 0$ at all coordinates vanishing in the given real solution. Together with the displayed equations they define a nonempty affine space over $\mathbb{Q}$. Gaussian elimination over $\mathbb{Q}$ gives a rational particular solution and a basis over $\mathbb{Q}$ for its direction space. Thus its rational points are dense in its real points. Approximate the given point closely enough to keep all of the finitely many remaining coordinates y_i strictly positive. This proves the first assertion, including the case where all y_i vanish or the affine space has dimension zero.

For the divisor assertion, choose a finite expression

$$D - G = \sum_{j=1}^v a_j \text{Div}(f_j), \quad a_j \in \mathbb{R}, \quad f_j \in k(X)^*.$$

List the prime divisors in the supports of D, G , and these finitely many principal divisors as $P_1, \dots, P_w$. Write $D = \sum d_i P_i$ and let $m_{ij} \in \mathbb{Z}$ be the coefficient of P_i in $\text{Div}(f_j)$. The coefficients of G and the a_j give a real solution of (8.1). A rational solution gives $G_{\mathbb{Q}} = \sum y_i P_i \geq 0$ with the same support and $D - G_{\mathbb{Q}} = \sum b_j \text{Div}(f_j)$.

Choose a positive integer N divisible by r and by the denominators of all y_i, b_j . Then

$$ND - NG_{\mathbb{Q}} = \text{Div} \left(\prod_j f_j^{Nb_j} \right).$$

The effective integral divisor $NG_{\mathbb{Q}}$ is Cartier, since ND is Cartier and their difference is principal. It defines a nonzero section of $\mathcal{O}_X(ND)$. Write $N = mr$. No $\mathbb{Q}$ -factoriality is needed: all equalities were equalities of Weil divisors on X . $\square$

Proof of Corollary 1.2. Hashizume's reduction [20, Theorem 1.4] takes smooth canonical nonvanishing in dimension four, supplied by Theorem 1.1, and gives nonvanishing for projective lc pairs with effective real boundaries and pseudo-effective real Cartier adjoints in dimensions at most four. Applied to the given rational lc pair, it produces $D \sim_{\mathbb{R}} G \geq 0$ on the original normal variety; nefness implies pseudo-effectivity. Lemma 8.1 gives a nonzero section of mrD for some positive integer m , with the prescribed index r . $\square$

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